TripAdvisor Review Analytics Report for Bvlgari Resort Bali: Data, Methodology, Results, and Future Decision-Support Extensions

Abstract

This report specifies a reproducible analytical framework for transforming online hotel reviews into structured sentiment evidence, and it presents the main descriptive outputs produced by that framework. The project uses a prepared TripAdvisor review corpus for Bvlgari Resort Bali and implements a staged R workflow for data validation, text preprocessing, lexicon-based sentiment scoring, emotion extraction, aspect-rating analysis, annual review profiling, and temporal visualization. The methodological design follows a single-case digital trace research strategy: one luxury resort is treated as a bounded case, while reviews are treated as naturally occurring electronic word-of-mouth records of guest experience (Filieri et al., 2015; Litvin, 2019). The final analytical dataset contains 762 TripAdvisor reviews from October 2006 to May 2026, including star ratings, review dates, stay dates, reviewer location, contribution counts, optional aspect ratings, cleaned review text, negation-adjusted Syuzhet sentiment scores, negation-adjusted AFINN sentiment scores, and NRC emotion scores.

The implemented framework supports academic analysis of the relationship between numerical ratings, textual sentiment, emotion categories, service-quality dimensions, and time-based review patterns. It now also produces statistical period-monitoring outputs that compare review periods with prior history using length-normalized sentiment, medians, trimmed means, and robust median absolute deviation indicators. A management and investment decision layer is not currently implemented in the project code; it is presented as a future development path that would build on the existing analytical outputs.

The paper also defines the study's limitations. Lexicon-based sentiment scoring is transparent and replicable, but it can misread sarcasm, complex negation, multilingual expressions, cultural context, and domain-specific meanings (Liu, 2020). TripAdvisor reviews are not a statistically representative sample of all guests, and observed sentiment is mediated by platform behavior, reviewer motivation, review timing, and unobserved operational conditions (Filieri et al., 2015; Litvin, 2019). For these reasons, the implemented methodology emphasizes triangulation across ratings, sentiment scores, aspect ratings, review text, and temporal patterns before any future management or investment conclusions are drawn.

Keywords: TripAdvisor, sentiment analysis, hotel reviews, luxury resort, Bvlgari Resort Bali, electronic word of mouth, AFINN, Syuzhet, NRC Emotion Lexicon, service quality, annual review profile, temporal analysis, future decision support.

1. Introduction

Online reviews have become a major source of evidence for tourism and hospitality research because they record guest evaluations in a natural digital setting and influence traveler trust, recommendation adoption, and word of mouth (Filieri et al., 2015). In luxury hospitality, reviews are especially valuable because guest satisfaction is shaped not only by functional quality but also by emotional experience, perceived exclusivity, value, personalization, aesthetics, privacy, and service recovery; hotel satisfaction and loyalty research has long emphasized the importance of service quality and image in shaping guest behavior (Kandampully & Suhartanto, 2000). A five-star property may maintain a high average rating while still accumulating text-based signs of dissatisfaction about value, maintenance, dining, staff coordination, or expectation mismatch. Conversely, a guest may give a modest numerical rating while writing positively about specific aspects of the experience. For this reason, rating-only analysis is methodologically incomplete.

This project develops a sentiment-analysis methodology for Bvlgari Resort Bali using TripAdvisor review data. The implemented objective is not merely to classify reviews as positive or negative; it is to create a transparent analytical corpus that can support rating-sentiment comparison, emotion profiling, and temporal interpretation. The methodology is inspired by tourism sentiment studies that combine digital review corpora, text cleaning, lexicon-based scoring, temporal interpretation, and qualitative contextualization (Adnyana et al., 2026; Liu, 2020). The attached reference paper on informal language and cultural appreciation in TripAdvisor reviews of Balinese Ayam Betutu demonstrates a rigorous academic pattern that is also relevant here: define a bounded tourism object, analyze all accessible reviews in the selected corpus, preprocess text transparently, apply lexicon-based sentiment methods, visualize temporal patterns, interpret discrepancies between ratings and text, and state limitations clearly (Adnyana et al., 2026).

The present analytical framework extends that type of approach to a luxury-hotel case. The current codebase produces cleaned data, scored sentiment outputs, annual review-profile summaries, aspect-rating diagnostics, and visual trend artifacts. A further decision-support layer is proposed as future work: rather than stopping at descriptive sentiment results, a future version could formalize what should happen when the rating or sentiment of reviews drifts away from a baseline over a number of periods. That future extension would require baseline definition, period aggregation, drift thresholds, confidence rules, and action pathways that are not yet coded in the repository.

2. Research Purpose and Methodological Objectives

The purpose of the study is to design and document a reproducible methodology for analyzing TripAdvisor reviews of Bvlgari Resort Bali. The implemented workflow produces research-ready sentiment, emotion, rating, and visualization outputs. Operational and investment decision rules are discussed only as future extensions that could be built on top of these outputs.

The methodological objectives are:

- To validate and standardize a prepared TripAdvisor review dataset for a single luxury hotel case.
- To preprocess guest review text into a clean corpus suitable for tokenization and lexicon-based sentiment analysis.
- To score review narratives using multiple lexicon-based methods, including Syuzhet, AFINN, and the NRC emotion categories.
- To compare textual sentiment with TripAdvisor star ratings and optional aspect ratings.
- To visualize sentiment distribution, rating-sentiment alignment, emotion profiles, word-level sentiment terms, and time-based sentiment trends.
- To identify how the implemented outputs could support a future early-warning framework for detecting rating and sentiment drift across periods.
- To outline, as future work, how persistent drift patterns could be mapped to management and investment decision categories.

The methodology is therefore primarily analytical. It explains how the data are processed and how the resulting indicators should be interpreted responsibly. Prescriptive management and investment rules are treated as a recommended extension, not as a feature currently implemented in the R workflow.

3. Research Design

3.1 Digital Trace Single-Case Study

The study uses a single-case digital trace design. Bvlgari Resort Bali is treated as the focal case, and TripAdvisor reviews are treated as digital traces of guest experience and electronic word of mouth (Filieri et al., 2015). A single-case design is appropriate because luxury resorts are highly context-dependent: the brand promise, location, villa format, service model, butler experience, food and beverage offer, price point, and guest expectations are all specific to the property. Cultural differences can also shape how travelers write and interpret TripAdvisor hotel reviews, which makes context-specific interpretation important (Litvin, 2019). A multi-hotel design would support broader generalization, but it would weaken the property-specific interpretation needed for any later decision-support extension.

3.2 Mixed Quantitative and Qualitative Logic

The project combines quantitative and qualitative logic. The quantitative component includes star ratings, aspect ratings, sentiment scores, emotion counts, review counts, period averages, medians, and trend indicators. The qualitative component remains embedded in the review text: low-scoring periods, outlier reviews, and divergent rating-sentiment cases must be interpreted by reading the guest narratives. This mixed logic follows the caution in sentiment-analysis research that automated scores should support interpretation rather than replace close reading of context (Adnyana et al., 2026; Liu, 2020).

3.3 Reproducible Computational Workflow

The workflow is implemented in R through separate scripts:

Script	Methodological role
`01_data_import.R`	Validates that the prepared raw review CSV exists and contains required fields.
`02_cleaning.R`	Loads the raw data, standardizes columns, cleans text, tokenizes reviews, removes stopwords, and saves cleaned outputs.
`03_sentiment_analysis.R`	Scores cleaned text using Syuzhet and AFINN, classifies review sentiment, extracts NRC emotion categories, and saves the scored dataset.
`04_visualization.R`	Produces sentiment distribution, rating-sentiment, emotion, word cloud, and temporal trend visualizations.
`05_aspect_text_analysis.R`	Links structured aspect ratings to review text by comparing low-aspect and high-aspect reviews, producing key-term tables, mismatch tables, qualitative examples, and aspect-text visualizations.
`scripts/data_config.R`	Centralizes file paths for raw data, cleaned data, sentiment scores, and figures.
`scripts/helpers.R`	Provides reusable helper functions for text cleaning, stopword removal, slang normalization, and source-column standardization.

This modular design improves transparency because each methodological stage can be inspected, rerun, and modified independently.

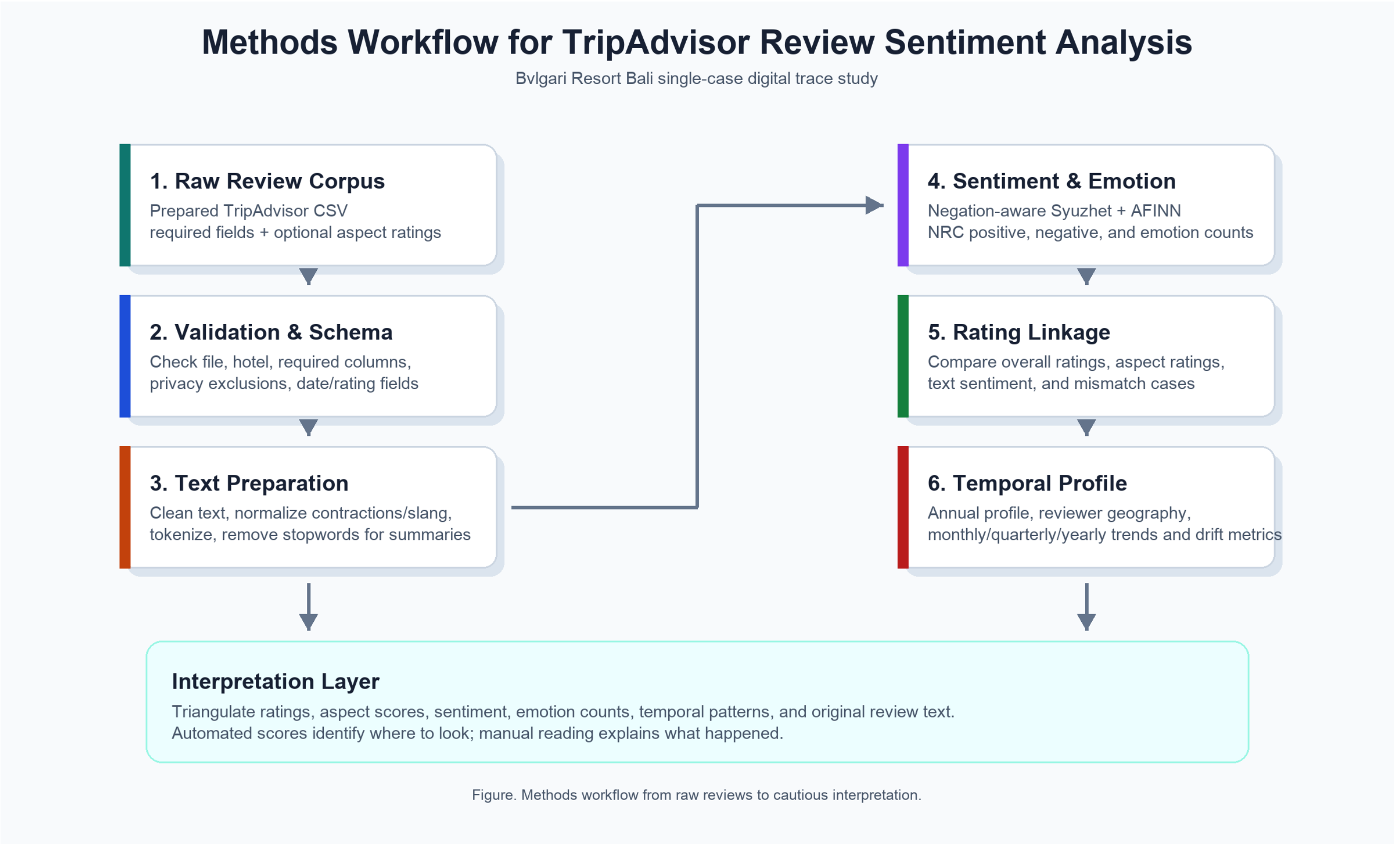

Figure 1. Methods workflow from raw TripAdvisor reviews to cautious interpretation.

The current repository also tracks the principal analytical outputs produced by these scripts. This is important for reproducibility because a reader can inspect both the source code and the generated evidence tables without rerunning the full workflow.

Output group	Key files	Analytical role
Cleaned datasets	`data/cleaned/hotel_cleaned_reviews.csv`, `data/cleaned/hotel_cleaned_tokens.csv`, `data/cleaned/hotel_sentiment_scores.csv`	Preserve the cleaned text, tokenized words, sentiment scores, emotion counts, and retained non-identifying metadata used in later analysis.
Temporal and geographic summaries	`output/reports/annual_review_profile.csv`, `output/reports/sentiment_period_summary.csv`	Summarize year-level review volume, reviewer geography, rating distributions, and period-level sentiment or rating drift metrics.
Aspect-rating summaries	`output/reports/aspect_rating_summary.csv`, `output/reports/high_overall_low_aspect_reviews.csv`	Compare structured aspect ratings and identify high-overall-rating reviews that still contain low aspect scores.
Aspect-text diagnostics	`output/reports/aspect_text_alignment.csv`, `output/reports/aspect_text_band_summary.csv`, `output/reports/aspect_text_key_terms.csv`, `output/reports/aspect_text_key_phrases.csv`, `output/reports/aspect_text_mismatches.csv`, `output/reports/aspect_qualitative_examples.csv`	Link aspect ratings to review text, extract low-aspect terms and phrases, identify signal mismatches, and support qualitative reading.
Figures	`output/figures/*.png`	Provide visual checks of sentiment distribution, rating-sentiment alignment, aspect ratings, aspect-text signals, emotion profiles, word clouds, temporal trends, and drift monitoring.

4. Case Selection

Bvlgari Resort Bali is selected as a luxury-hotel case because its market position depends heavily on intangible experience attributes: exclusivity, privacy, aesthetics, personalized service, food and beverage quality, spa and leisure amenities, room and villa quality, and perceived value. These dimensions are well suited to review-based analysis because guests often describe them in narrative form, and hospitality research links service quality, atmosphere, and related experience dimensions to revisit and satisfaction outcomes (Bichler et al., 2021; Kandampully & Suhartanto, 2000).

The single-property focus also supports future management relevance. A general hotel sentiment model can describe broad industry patterns, but managers need property-specific evidence. For example, an observed decline in sentiment could imply different follow-up questions depending on whether the associated text concerns butler service, dining price, villa maintenance, cleanliness, privacy, buggy transport, beach access, or value perception. The current project surfaces these analytical signals; it does not yet automate managerial recommendations.

5. Data Source and Corpus Construction

5.1 Raw Dataset

The project uses a prepared TripAdvisor review CSV located at `data/raw/reviews.csv`. The import script validates that the file exists and contains the minimum required columns needed for the core workflow:

- `review_id`
- `hotel_name`
- `title`
- `review_text`
- `rating`
- `review_date`
- `stay_date`
- `trip_type`

This list is the minimum import contract, not the full analytical schema. The current dataset also contains optional fields that are used when available, including `reviewer_location`, `reviewer_contributions`, `insider_tip`, `source`, and the six structured aspect-rating columns for value, rooms, location, cleanliness, service, and sleep quality. The `reviewer_location` column is present in the current corpus and is retained for aggregate geographic profiling, but it is not required because the cleaning, sentiment, and aspect-rating workflow can still run on a prepared review file that lacks reviewer geography.

The current raw dataset contains 762 rows and 20 source columns after removing direct reviewer names and source-response metadata while retaining reviewer location for aggregate geographic profiling. All rows are associated with Bvlgari Resort Bali and have non-empty review text. The final sentiment-scored dataset contains 762 rows and 38 columns.

5.2 Temporal Coverage

The review-date range in the current dataset is October 2006 to May 18, 2026. The stay-date range is September 2006 to May 2026. This long time span enables longitudinal analysis, but it also creates methodological challenges. TripAdvisor platform design, guest demographics, hotel operations, travel patterns, and global tourism conditions changed substantially over this period. Because TripAdvisor reviews are shaped by platform trust, reviewer behavior, and cultural context, long-term trend interpretation must consider context and should not treat all periods as directly comparable (Filieri et al., 2015; Litvin, 2019).

5.3 Data Fields

The raw corpus includes both narrative and structured fields:

Field group	Examples	Methodological use
Identification	`review_id`, `hotel_name`, `source`	Traceability and data validation.
Review content	`title`, `review_text`	Text cleaning, tokenization, sentiment analysis, qualitative interpretation.
Rating fields	`rating`, `value_rating`, `rooms_rating`, `location_rating`, `cleanliness_rating`, `service_rating`, `sleep_quality_rating`	Rating-sentiment comparison and aspect-level diagnosis.
Time fields	`review_date`, `review_date_raw`, `stay_date`	Monthly, quarterly, yearly, and rolling trend analysis.
Reviewer context	`reviewer_location`, `reviewer_contributions`, `trip_type`	Aggregate geographic profiling and optional segmentation without retaining reviewer names.
Operational hints	`insider_tip`	Supplementary guest context when present.

Direct reviewer identifiers, including bare `author`, `reviewer`, `reviewer_name`, profile IDs, and profile URLs, are removed from the tracked analysis datasets because they are not required for the implemented aggregate analysis. The `reviewer_location` field is retained because the annual profile uses it only in aggregate form. The validation and standardization code treats `reviewer_location` as the only reviewer-geography field allowed into the analysis schema; non-whitelisted variants such as `user_location`, `author_location`, `member_location`, and `profile_location` are rejected or dropped. Common source-column aliases such as `Reviewer Location` and `reviewerLocation` are canonicalized to `reviewer_location` so the annual profile remains stable across export formats. Academic and managerial outputs should avoid person-level reviewer disclosure unless explicit permission exists.

5.4 Corpus Summary

The current scored dataset has the following descriptive structure:

Metric	Value
Total reviews	762
Hotel name	Bvlgari Resort Bali
Review-date range	2006-10-01 to 2026-05-18
Stay-date range	2006-09-01 to 2026-05-01
Mean star rating	4.56
Median star rating	5.00
5-star reviews	596
4-star reviews	77
3-star reviews	38
2-star reviews	22
1-star reviews	29
Positive sentiment labels	733
Neutral sentiment labels	2
Negative sentiment labels	27
Mean AFINN score	21.13
Median AFINN score	17.00
Mean Syuzhet score	8.60
Median Syuzhet score	6.62

These figures show a strongly positive and high-rating corpus. This high skew is typical of many luxury-hotel review datasets and makes methodological caution necessary. Mean values alone can hide deterioration in a small but important subset of reviews. Therefore, the current monitoring outputs include median and robust summaries, while future decision-support work should add richer low-rating share, negative-share, and aspect-specific trigger logic.

5.5 Annual Review Profile: Temporal, Geographical, and Rating Analysis

The workflow writes `output/reports/annual_review_profile.csv` to summarize the corpus by year. This table follows the logic of the reference paper's annual profile by combining temporal coverage, reviewer geography, rating average, and rating distribution. The geography column uses `reviewer_country`, a normalized country field derived from the prepared dataset's self-reported `reviewer_location` field. Because the field is self-reported and incomplete, missing values are labeled `Unknown`, and only countries with at least two reviews in a year are listed. To keep the report readable, the table lists the four most frequent countries at the threshold and reports how many additional qualifying country groups were omitted from the printed cell.

Year	Reviews	Reviewer countries, n>=2	Avg rating	5	4	3	2	1
2026	72	Unknown (66), United States (2)	5.00	72	0	0	0	0
2025	18	Unknown (12)	4.94	17	1	0	0	0
2024	17	Unknown (8), United States (4)	4.47	14	1	0	0	2
2023	20	Unknown (6), India (2), United Kingdom (2), United States (2)	4.95	19	1	0	0	0
2022	25	Indonesia (9), Unknown (5), Singapore (4), India (2)	4.52	21	1	0	1	2
2021	10	Indonesia (6), Unknown (2)	4.80	9	0	1	0	0
2020	14	Australia (2), United Arab Emirates (2)	4.93	13	1	0	0	0
2019	60	United States (16), Unknown (11), Australia (4), China (4), plus 4 other country groups	4.65	50	5	2	0	3
2018	76	Unknown (14), China (11), United States (9), Indonesia (8), plus 9 other country groups	4.68	64	5	3	3	1
2017	93	Unknown (21), Indonesia (9), United States (9), Australia (7), plus 10 other country groups	4.72	77	10	3	2	1

Year	Reviews	Reviewer countries, n>=2	Avg rating	5	4	3	2	1
2016	89	Unknown (13), Australia (12), Indonesia (11), United States (11), plus 6 other country groups	4.48	67	8	8	2	4
2015	58	Unknown (12), United States (9), Indonesia (8), Australia (6), plus 4 other country groups	4.50	41	9	5	2	1
2014	41	Singapore (7), Australia (6), United States (4), China (3), plus 5 other country groups	4.05	24	8	1	3	5
2013	48	Australia (11), United States (8), Singapore (7), Indonesia (3), plus 5 other country groups	4.19	32	4	5	3	4
2012	48	Australia (9), United States (9), Indonesia (7), China (5), plus 4 other country groups	4.29	31	7	6	1	3
2011	23	Australia (6), United States (3), Hong Kong (2), Singapore (2)	4.22	15	3	2	1	2
2010	12	Singapore (3), United States (3), Hong Kong (2)	4.08	6	3	2	0	1
2009	21	United States (4), Singapore (3), India (2), United Kingdom (2)	4.48	14	5	0	2	0
2008	10	Australia (2)	4.40	6	3	0	1	0
2007	4	No country reached the minimum review count	3.75	1	2	0	1	0
2006	3	United States (2)	5.00	3	0	0	0	0
Total	762	Unknown (178), United States (98), Australia (73), Indonesia (70), plus 34 other country groups	4.56	596	77	38	22	29

The annual profile shows that the dataset is not evenly distributed across time. Review volume is highest in 2016, 2017, 2018, and the partial 2026 period. Years with very low volume, such as 2006 and 2007, should be interpreted cautiously because a small number of reviews can move the annual average sharply. This annual profiling approach follows the temporal-summary logic used in the reference TripAdvisor sentiment study (Adnyana et al., 2026). The rating distribution is also highly skewed: 596 of 762 reviews are five-star reviews, while 89 reviews are rated one to three stars. This skew supports the decision to examine low-rating shares, sentiment distributions, and text-level complaints rather than relying only on the overall mean rating.

The reviewer-country profile indicates broad international reach, with the United States, Australia, Indonesia, Singapore, China, the United Kingdom, and other markets appearing repeatedly in the table. However, 178 reviews have missing reviewer geography, and recent years contain a particularly high share of `Unknown` locations. Therefore, reviewer geography should be treated as contextual evidence about the visible reviewer base, not as a representative market-origin distribution.

The country sentiment chart adds a second geographic view. It summarizes the median length-normalized AFINN sentiment score for each reviewer country with at least five reviews, excluding `Unknown` locations. Median sentiment is preferable to mean sentiment here because reviewer countries have uneven review counts and individual reviews can be unusually positive or negative. The chart is therefore useful for comparing visible reviewer-country groups, but it should not be interpreted as a representative market-origin ranking.

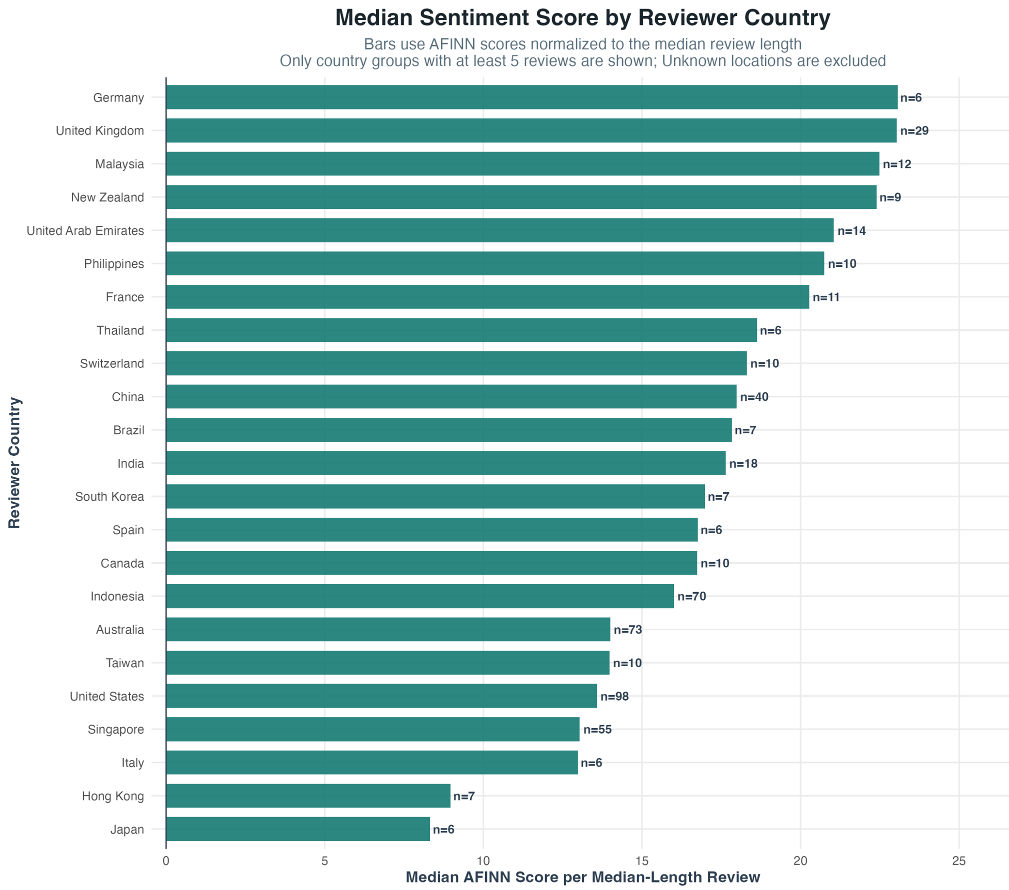

Figure 1b. Median length-normalized sentiment score by reviewer country.

5.6 Structured Aspect Rating Availability

The dataset also includes structured aspect ratings for value, rooms, location, cleanliness, service, and sleep quality. These fields are not merely supplementary metadata. They provide a middle layer between the overall star rating and the open-ended review text. Overall ratings indicate broad satisfaction, while aspect ratings show which part of the guest experience may be driving satisfaction or dissatisfaction.

Coverage varies by aspect because not every reviewer supplies every structured rating. Therefore, aspect analysis must report the number of available ratings before interpreting any aspect-level score:

Aspect	Rated reviews	Coverage
Value	339	44.5%
Rooms	322	42.3%
Location	337	44.2%
Cleanliness	327	42.9%
Service	440	57.7%
Sleep quality	272	35.7%

Substantive aspect findings are presented later in the analytical sequence, after the paper explains rating-sentiment alignment. This ordering prevents the corpus description section from drawing operational conclusions before the relevant method has been introduced.

6. Data Validation and Standardization

The first methodological stage is validation. `01_data_import.R` checks that the prepared CSV exists, has non-zero file size, includes required columns, contains rows with review text, identifies Bvlgari Resort Bali as the only analyzed property, and omits direct reviewer identifiers, forbidden source metadata, TripAdvisor source-disclosure text in trip type values, and embedded contact markers from any raw column. The only reviewer-location field allowed through this gate is `reviewer_location`, which is used for aggregate annual geography only. This step prevents downstream scripts from silently analyzing an incomplete, malformed, mixed-property, or privacy-inappropriate file.

The second stage is schema standardization. The helper function `standardize_hotel_reviews()` maps source-specific column names into a consistent internal structure:

- `review_id`
- `hotel_name`
- `title`
- `review_text`
- `rating`
- `review_date`

The function also preserves additional source fields, including aspect ratings and non-identifying reviewer context such as contribution counts. Direct reviewer identifiers and source-response metadata are excluded. This design is important for reproducibility because review exports may differ in column naming. A standardized schema allows the cleaning, scoring, and visualization scripts to remain stable even if the raw export format changes.

7. Text Preprocessing

7.1 Cleaning

Text preprocessing is implemented in `scripts/helpers.R` through the `clean_text()` function. The cleaning process:

- Converts text to lowercase.
- Removes HTML tags.
- Removes loose angle brackets, URLs, and web handles.
- Transliterates accented letters and Unicode punctuation to ASCII where possible.
- Replaces remaining non-ASCII characters and emojis with spaces so neighboring words do not get joined together.
- Removes punctuation and special symbols.
- Removes digits.
- Squishes repeated whitespace.

This procedure produces a simplified textual representation suitable for lexicon matching. The main advantage is transparency: the rule set is explicit and reproducible. The main disadvantage is loss of some linguistic nuance. For example, punctuation, capitalization, emojis, and non-English characters may carry sentiment information. This loss is acceptable for a first-stage lexicon workflow, but it should be acknowledged as a limitation.

7.2 Slang Normalization

The `normalize_slang()` function applies a small dictionary of common informal expressions, such as mapping elongated or abbreviated forms to standard English forms. Examples include:

- `sooo` to `so`
- `gooo+d` to `good`
- `looo+ve` to `love`
- `pls` to `please`
- `vry` to `very`

This step recognizes that TripAdvisor reviews often include informal language. However, the current dictionary is intentionally conservative. Expanding it should be done carefully and documented because aggressive normalization can erase culturally meaningful expressions.

The helper also standardizes straight and smart apostrophe contractions before sentiment scoring. This matters because expressions such as "don't recommend" or "didn't like" must be converted to explicit negation forms before punctuation and Unicode cleanup, otherwise the negation can be lost and positive words such as "recommend" can be scored incorrectly. Unicode punctuation is also treated as a word boundary rather than deleted, which prevents joined tokens from distorting sentiment scores and aspect key-term reports.

7.3 Tokenization

The cleaned review text is tokenized using `tidytext::unnest_tokens()`, creating one row per word. Tokenization supports word-frequency analysis, stopword removal, word clouds, and experience-word filtering. The final token file contains 48,563 token rows after preprocessing.

7.4 Stopword Removal

Stopwords are removed using the `tidytext` stopword list. This reduces the influence of high-frequency function words such as "the", "and", and "is". Stopword removal is appropriate for word-frequency visualization but should not be used blindly for every NLP task. Some function words can matter for negation, modality, or phrase-level meaning. Because this project uses lexicon-based word matching, stopword removal is primarily used for token-level summaries rather than for preserving syntactic structure, following tidy text principles for transparent token-level analysis (Silge & Robinson, 2016).

Figure 1. Guest experience word cloud after text cleaning and combined filtering.

In the word cloud, teal words mark positive or emotional language, gold words mark hotel-experience topics, muted red words mark negative or complaint-oriented terms, and purple words mark other frequent context terms.

8. Sentiment and Emotion Scoring

8.1 Lexicon-Based Sentiment Rationale

The study uses lexicon-based sentiment analysis because it is transparent, replicable, and suitable for educational settings. A lexicon approach maps words to preassigned sentiment or emotion values, a standard family of methods in sentiment-analysis research (Liu, 2020). Unlike black-box machine-learning models, lexicon results can be inspected and explained to non-technical stakeholders. This transparency would also be useful if a future management or investment decision-support layer were added.

The limitation is that lexicons may not fully capture sarcasm, idioms, multilingual content, domain-specific meanings, complex negation, or context-dependent sentiment (Liu, 2020). For instance, a word that is negative in a political context may be neutral in a hotel context. Cultural differences in review expression can also affect hotel-review interpretation (Litvin, 2019). The visualization script therefore builds the word cloud from a combined set of sentiment words, NRC emotion words, hotel-experience terms, and frequent context words while excluding broad filler terms.

8.2 Syuzhet Score

The Syuzhet score is calculated with the project's `score_negation_aware_sentiment()` helper using the Syuzhet lexicon. The helper scores each word, then reverses sentiment-bearing words that appear shortly after simple negators such as "not", "never", "without", and "cannot". This prevents phrases such as "cannot recommend" from being treated as positive only because "recommend" is a positive lexicon word. The resulting Syuzhet score provides a continuous sentiment estimate for each cleaned review, consistent with lexicon-based scoring principles described in sentiment-analysis literature (Liu, 2020). The project uses the Syuzhet score to classify reviews into:

- Positive: Syuzhet score greater than 0
- Negative: Syuzhet score less than 0
- Neutral: Syuzhet score equal to 0

This classification is simple and transparent, but it should not be overinterpreted. A very slightly positive score and a strongly positive score are both labeled positive, so the continuous score should be retained for analysis. In the aspect-text diagnostics, Syuzhet is also normalized to the dataset's median cleaned review length before aspect-level averaging so that longer reviews do not automatically receive larger aspect sentiment means.

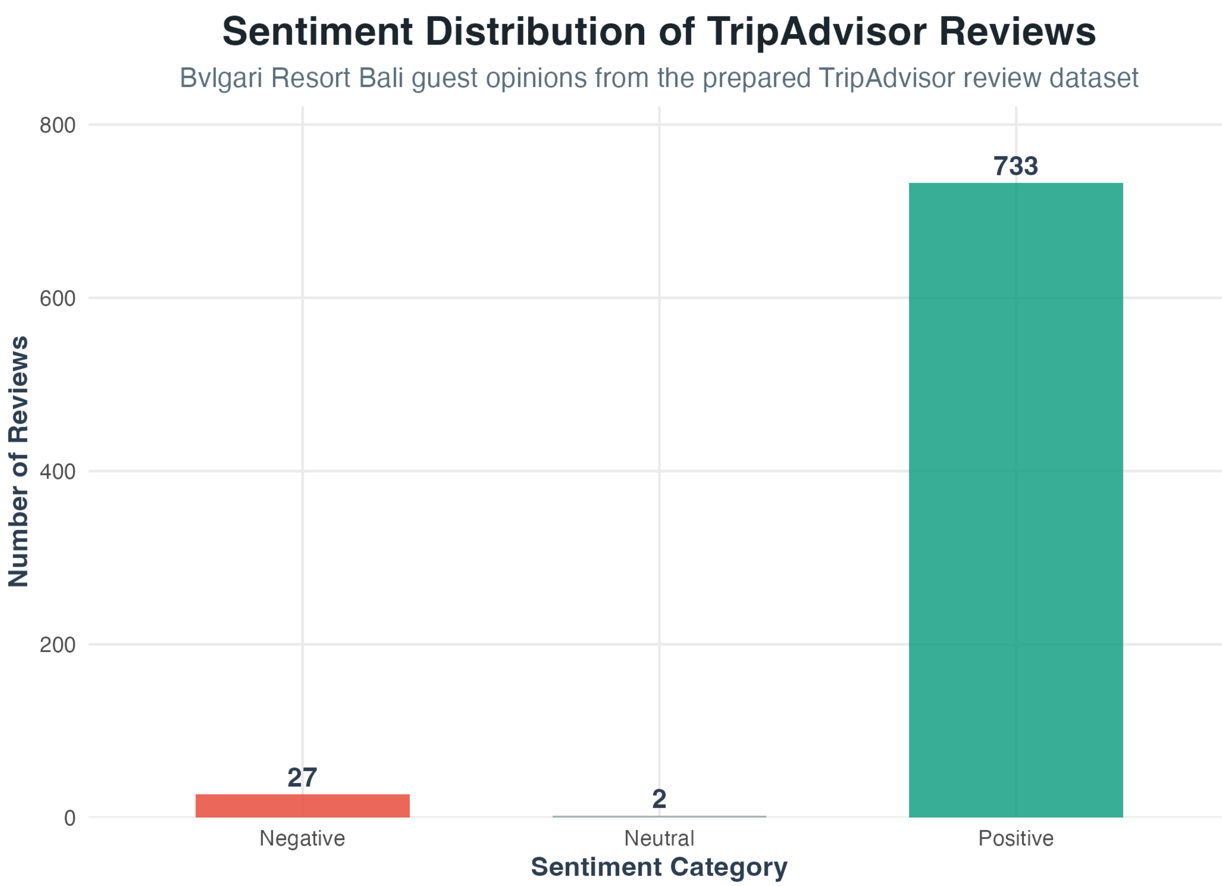

Figure 2. Distribution of positive, neutral, and negative review sentiment labels.

8.3 AFINN Score

The AFINN score is calculated with the same `score_negation_aware_sentiment()` helper using the AFINN lexicon. AFINN assigns integer valence values to sentiment-bearing words and sums them across text after the simple negation adjustment (Nielsen, 2011). In this project, AFINN is especially useful for:

- Rating-sentiment boxplots.
- Monthly, quarterly, and yearly trend charts, after normalizing AFINN to the median cleaned review length.
- Rolling sentiment averages, using the same length-normalized trend score.
- Prior-period comparisons, using the same length-normalized trend score.
- Aspect-text alignment diagnostics, using the same length-normalized score when comparing low and high aspect ratings.

Summed AFINN scores are sensitive to review length because longer reviews contain more sentiment words. This is not inherently wrong, because longer reviews may express more total evaluative content, but it should be considered when comparing short and long reviews. For that reason, the temporal charts calculate AFINN scaled to the dataset's median cleaned review length before averaging monthly, quarterly, and yearly periods, and the aspect-text diagnostics use the same median-length normalization for AFINN and Syuzhet aspect-level summaries. In the current corpus, the median cleaned review length is 112 words.

8.4 NRC Emotion Scores

The NRC emotion extraction is performed using `get_nrc_sentiment()`. The underlying NRC Emotion Lexicon is designed to associate words with positive and negative sentiment as well as basic emotion categories (Mohammad & Turney, 2013). It produces counts for the following emotion dimensions:

- anger
- anticipation
- disgust
- fear
- joy
- sadness
- surprise
- trust

It also produces positive and negative word counts. Emotion categories are useful for future operational interpretation because they distinguish different types of dissatisfaction. For example, anger may indicate service failure or perceived unfairness, disgust may indicate cleanliness or food-quality issues, fear may indicate safety or uncertainty, and sadness may indicate disappointment relative to expectations. These interpretations should be verified through the text rather than inferred from emotion labels alone.

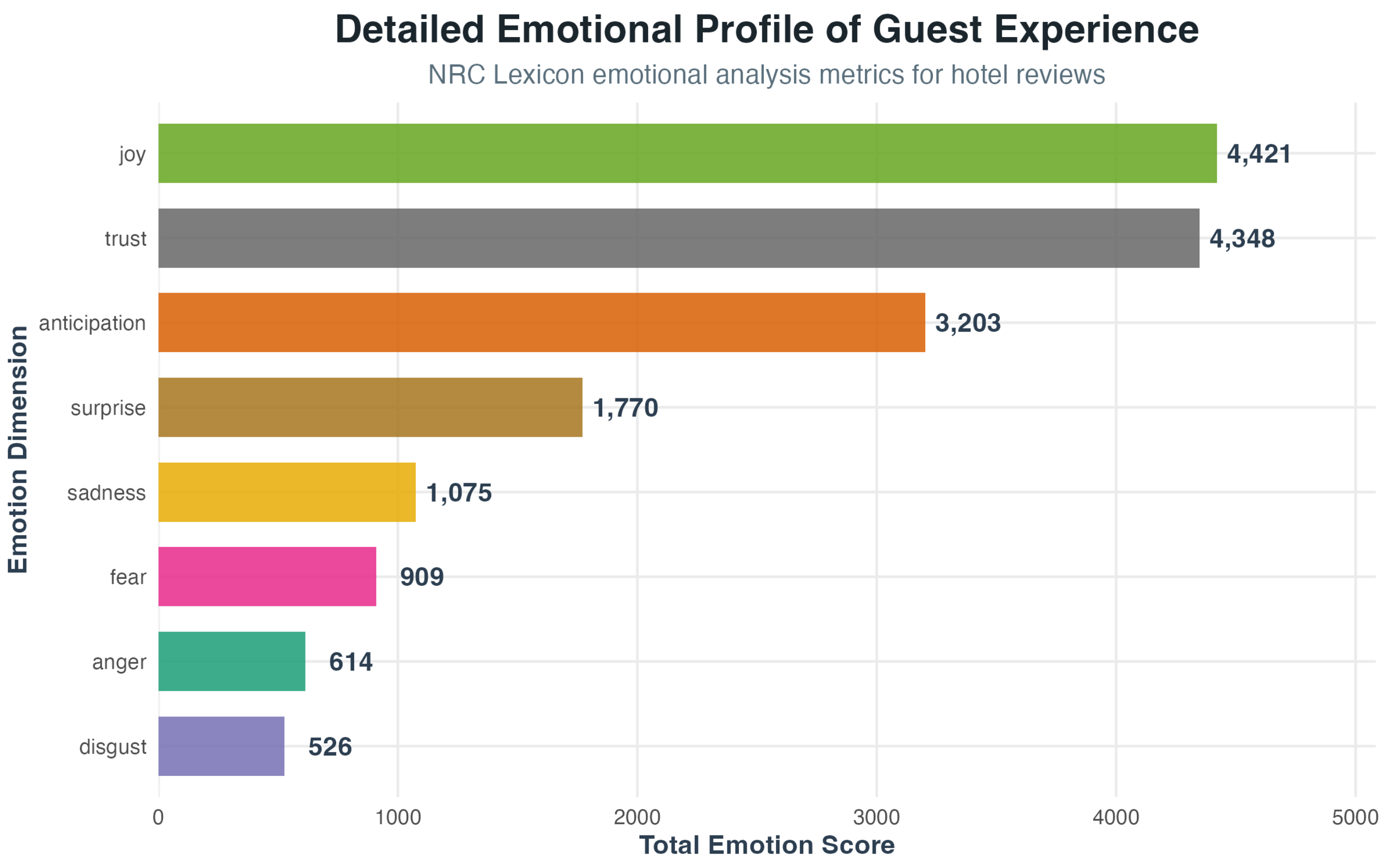

Figure 3. NRC emotion-category counts across the review corpus.

9. Rating-Sentiment Alignment

The methodology compares star ratings with textual sentiment because ratings and review narratives capture related but different constructs. Ratings are compressed numerical judgments. Text contains explanations, contradictions, and nuance.

The project produces a rating-sentiment boxplot that groups AFINN scores by TripAdvisor star rating. This supports three analytical checks:

- Convergent validity: higher star ratings should generally align with higher text sentiment.
- Divergence detection: high ratings with low sentiment may indicate polite rating behavior or mixed experiences.
- Severity detection: low ratings with strongly negative text may indicate service failures requiring deeper qualitative review.

For future management interpretation, rating-sentiment divergence may be more useful than average sentiment. A five-star review that contains negative comments about dining, value, or room maintenance could serve as an early warning before the numerical rating declines. The current project visualizes this relationship but does not yet implement automated warning logic.

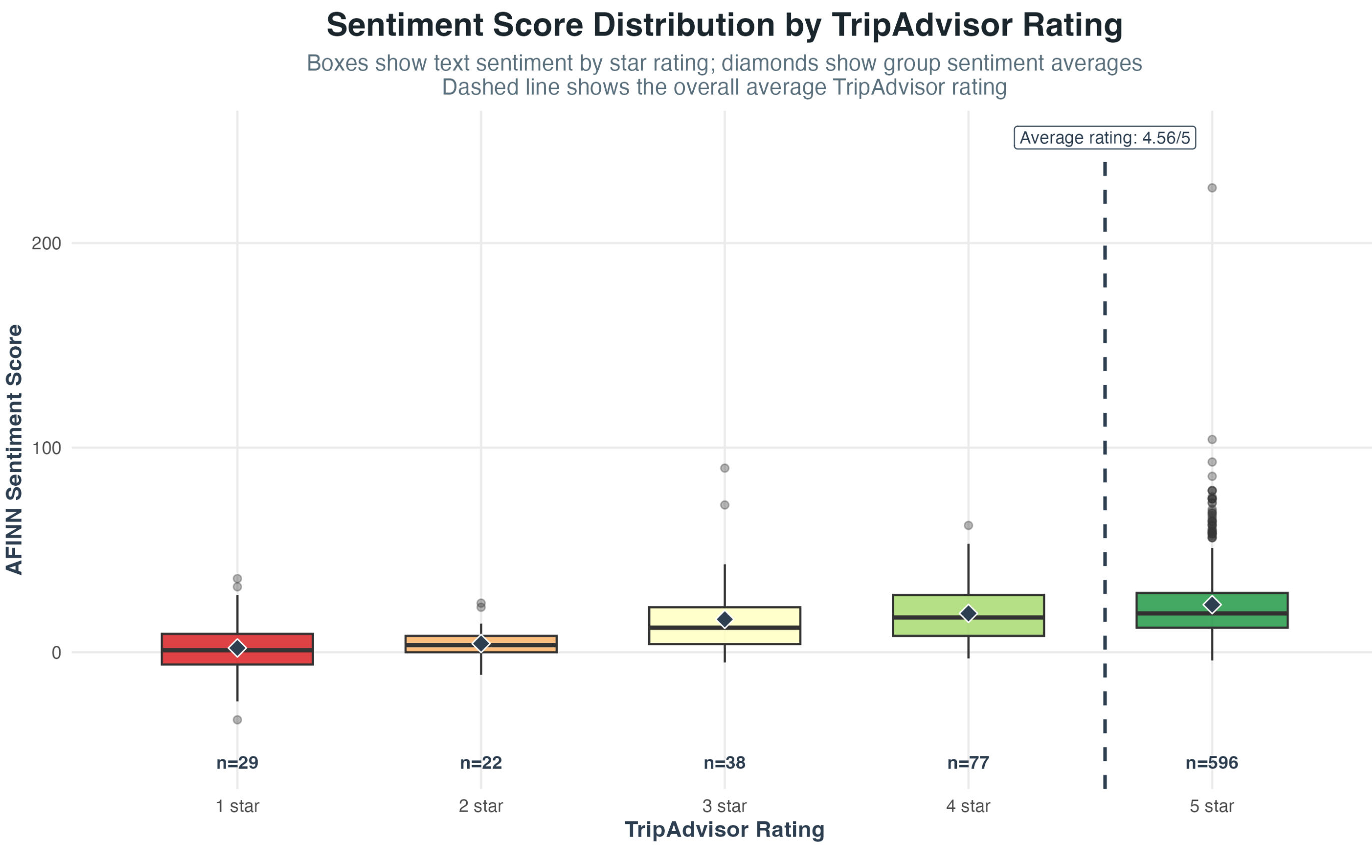

Figure 4. Distribution of AFINN text sentiment by TripAdvisor star rating.

10. Aspect-Rating Analysis

The corpus includes optional aspect ratings for value, rooms, location, cleanliness, service, and sleep quality. These aspect ratings provide a bridge between unstructured text and operational departments, which is relevant because hospitality service-quality research treats experience dimensions as distinct drivers of guest satisfaction and revisit behavior (Bichler et al., 2021; Kandampully & Suhartanto, 2000).

The implemented visualization workflow now treats aspect ratings as part of the core analysis rather than as a future extension. It generates an aspect summary table, a low-score table, a yearly aspect-rating heatmap, and a list of reviews where the overall rating is high but at least one aspect rating is low.

Aspect ratings should be interpreted with two statistics rather than one. The mean indicates the central tendency, while the low-score share indicates the proportion of guests who rated that aspect from 1 to 3 stars. Low-score share is important because a luxury property can maintain a high average while still accumulating a meaningful minority of weak aspect evaluations.

Aspect	Valid ratings	Coverage	Mean	Median	Low-score share, 1-3
Value	339	44.5%	4.06	4.00	25.1%
Rooms	322	42.3%	4.52	5.00	12.4%
Location	337	44.2%	4.56	5.00	11.3%
Cleanliness	327	42.9%	4.57	5.00	10.7%
Service	440	57.7%	4.67	5.00	7.7%
Sleep quality	272	35.7%	4.55	5.00	10.7%

In this dataset, value is the weakest structured dimension: it has the lowest mean rating and the highest low-score share. Methodologically, this does not automatically imply a pricing problem, but it does identify value perception as an important diagnostic area for qualitative review. Service is the strongest structured dimension, with the highest mean and the lowest low-score share.

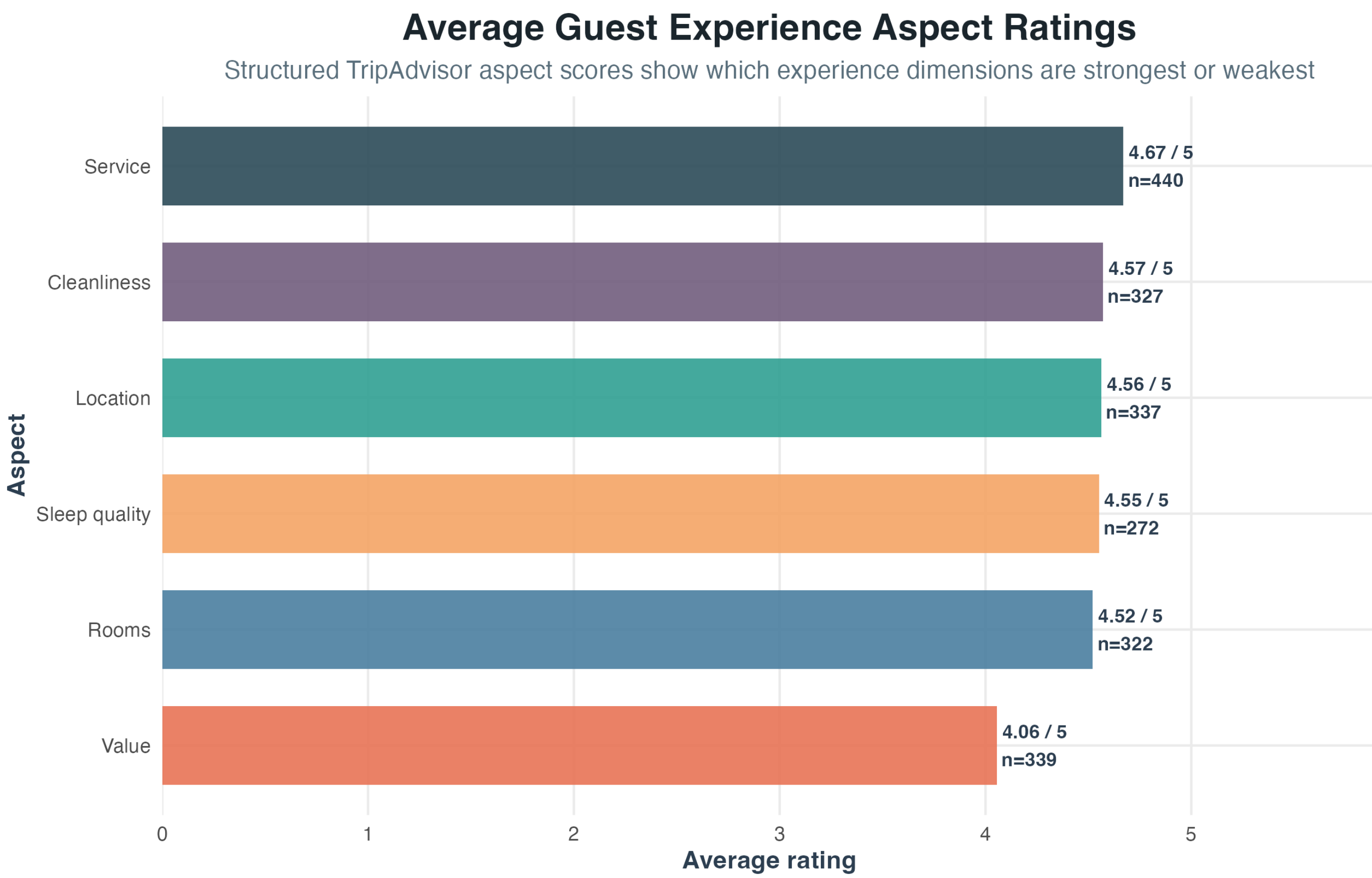

Figure 5. Mean structured aspect ratings by guest-experience dimension.

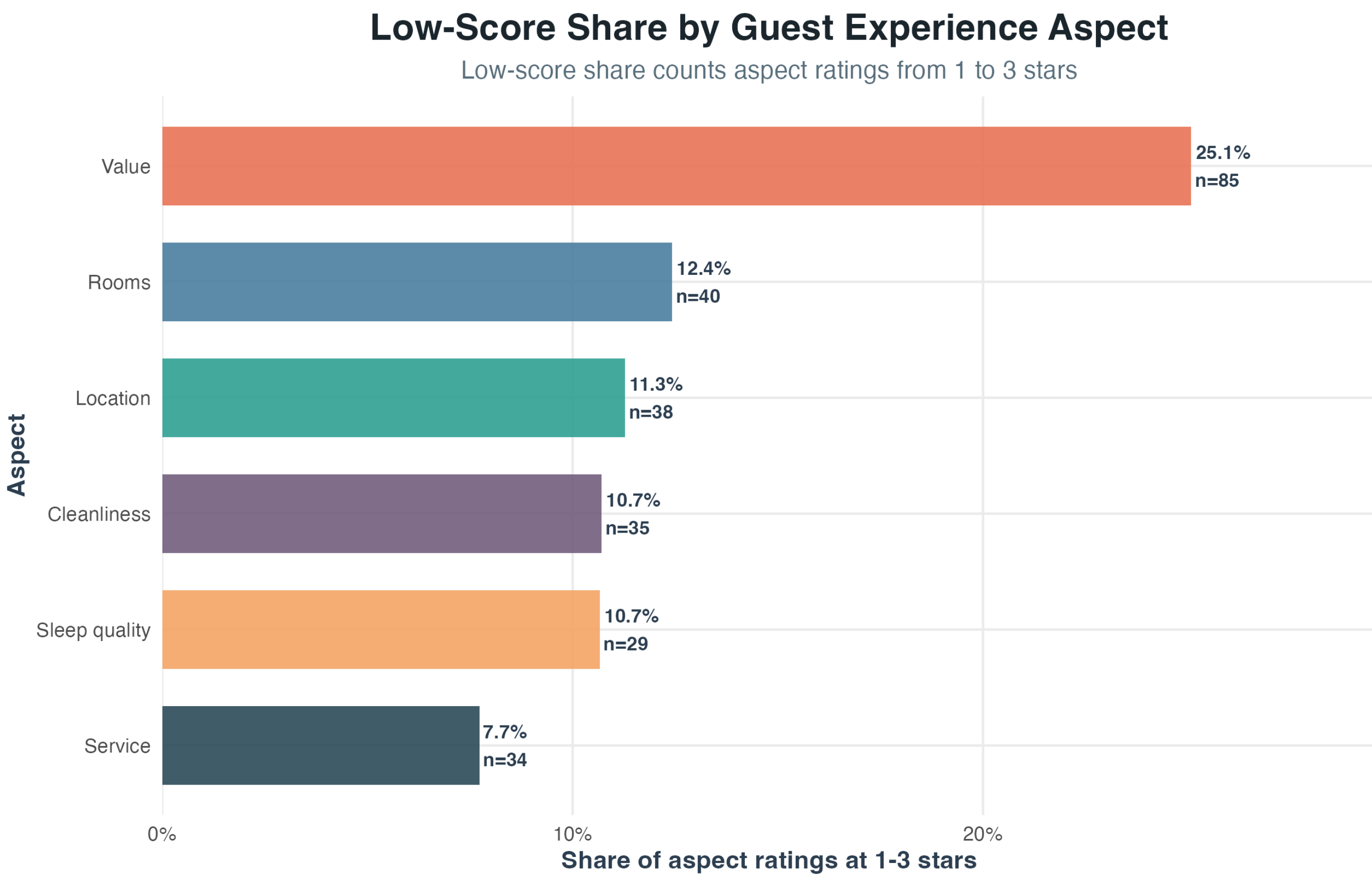

Figure 6. Share of low structured aspect ratings by guest-experience dimension.

The project also generates a review-level table for cases where the overall rating is high, but at least one aspect rating is low. In the current corpus, 60 reviews have an overall rating of 4 or 5 while reporting at least one aspect score of 3 or lower. These reviews are analytically important because they show mostly satisfied guests who still signaled specific weaknesses. For example, a five-star review with a low value score may reflect admiration for the property but hesitation about price fairness, inclusions, or expectation setting.

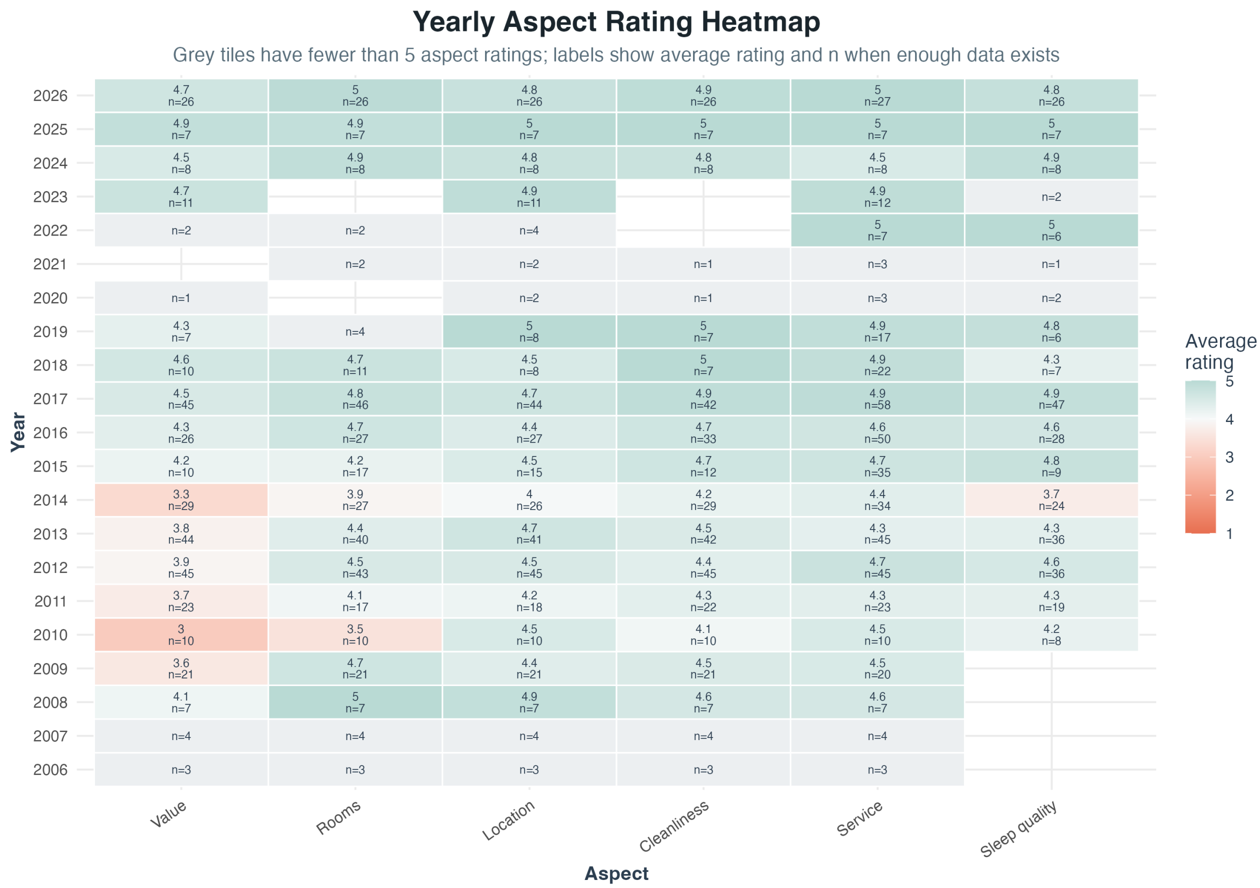

Figure 7. Yearly structured aspect-rating heatmap with available rating counts.

For future management interpretation, aspect ratings should be mapped to likely operational domains:

Aspect signal	Likely diagnostic domain
Value weakness	Price fairness, inclusions, package design, communication of luxury value.
Rooms weakness	Villa condition, lighting, bedding, maintenance, renovation priority.
Cleanliness weakness	Housekeeping quality assurance and inspection process.
Service weakness	Staffing, training, responsiveness, personalization, service recovery.
Sleep-quality weakness	Noise, bedding, room environment, privacy, night operations.
Location weakness	Transportation, accessibility, arrival experience, expectation setting.

This mapping is interpretive rather than causal. Any future action should be grounded in the review text and, where possible, triangulated with operational records.

The current workflow now implements that linkage in `05_aspect_text_analysis.R`. The script treats aspect ratings as weak review-level labels and compares the text of low-aspect reviews against high-aspect reviews. This produces five additional evidence types:

- Aspect-conditioned word and phrase tables that show terms appearing unusually often in low-scoring aspect reviews.
- Aspect-specific sentiment summaries that compare text sentiment, negative-text share, and emotion counts by structured aspect.
- Aspect-text mismatch tables for cases such as high overall ratings with low aspect ratings, or high aspect ratings with negative text.
- Qualitative review examples that support manual reading of low-aspect cases.
- Visualizations including aspect sentiment boxplots, low-score word and phrase charts, a negative-term heatmap, and mismatch count charts.

The current aspect-text outputs show why this middle layer is useful. `aspect_text_band_summary.csv` shows that low-aspect reviews have substantially lower median-length-normalized AFINN sentiment and higher negative-text shares than high-aspect reviews across every structured dimension:

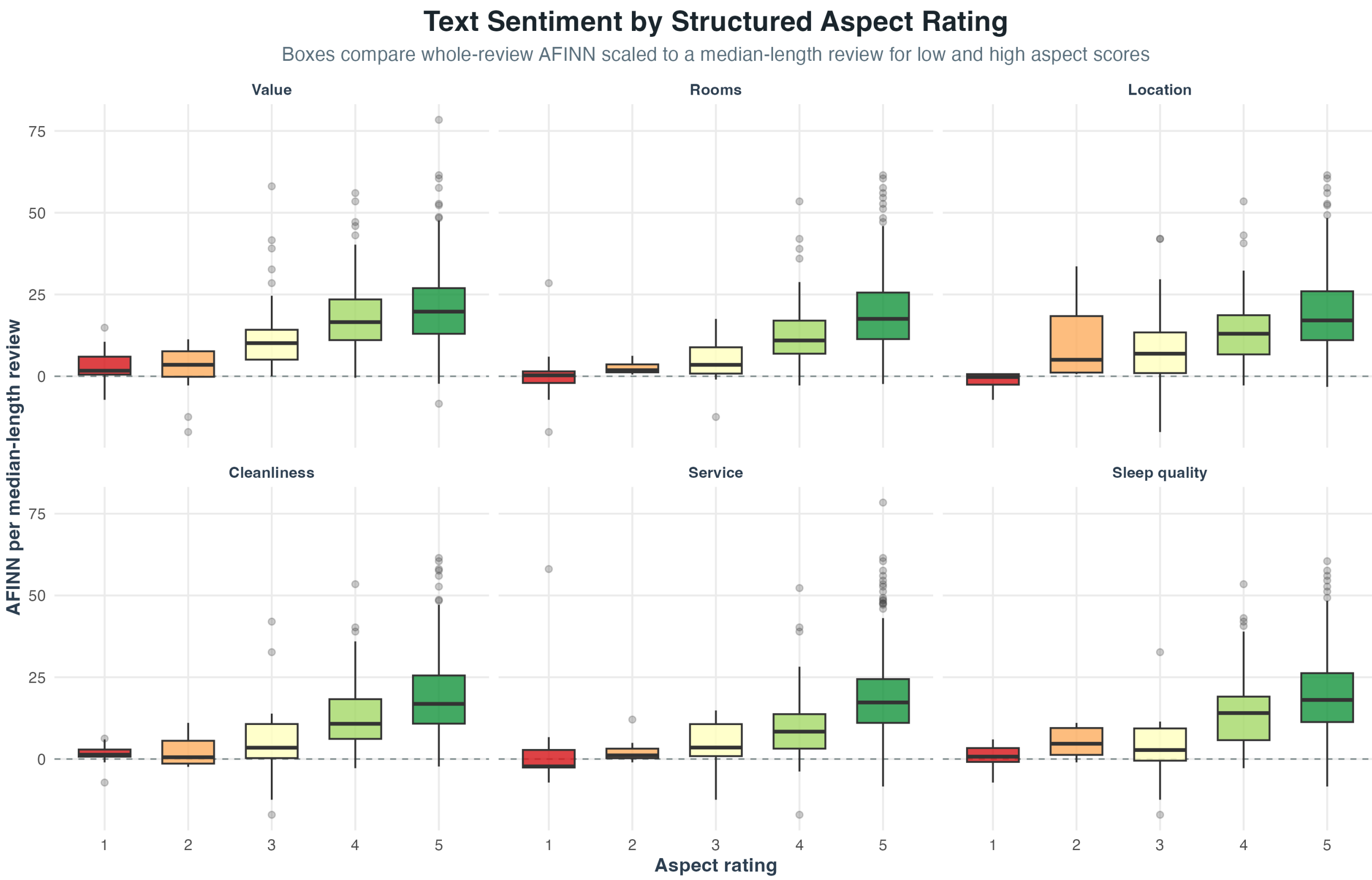

Figure 8. Whole-review text sentiment distribution by structured aspect-rating level.

Aspect	High-aspect mean AFINN	Low-aspect mean AFINN	High-aspect negative-text share	Low-aspect negative-text share
Value	20.72	7.96	1.6%	10.6%
Rooms	18.63	3.25	2.5%	15.0%
Location	18.28	8.34	2.3%	18.4%
Cleanliness	18.55	3.84	1.7%	22.9%
Service	18.30	4.25	2.2%	23.5%
Sleep quality	19.50	2.86	2.1%	27.6%

The low-aspect key-term and key-phrase tables also surface interpretable weak-label signals. For example, value-related low-score reviews emphasize terms and phrases such as `overpriced`, `difficult`, `not worth`, `not cleaned`, and `not return`. These terms should not be treated as automatic causal proof, but they are useful pointers for manual review, consistent with the broader sentiment-analysis caution that lexical evidence requires contextual interpretation (Liu, 2020).

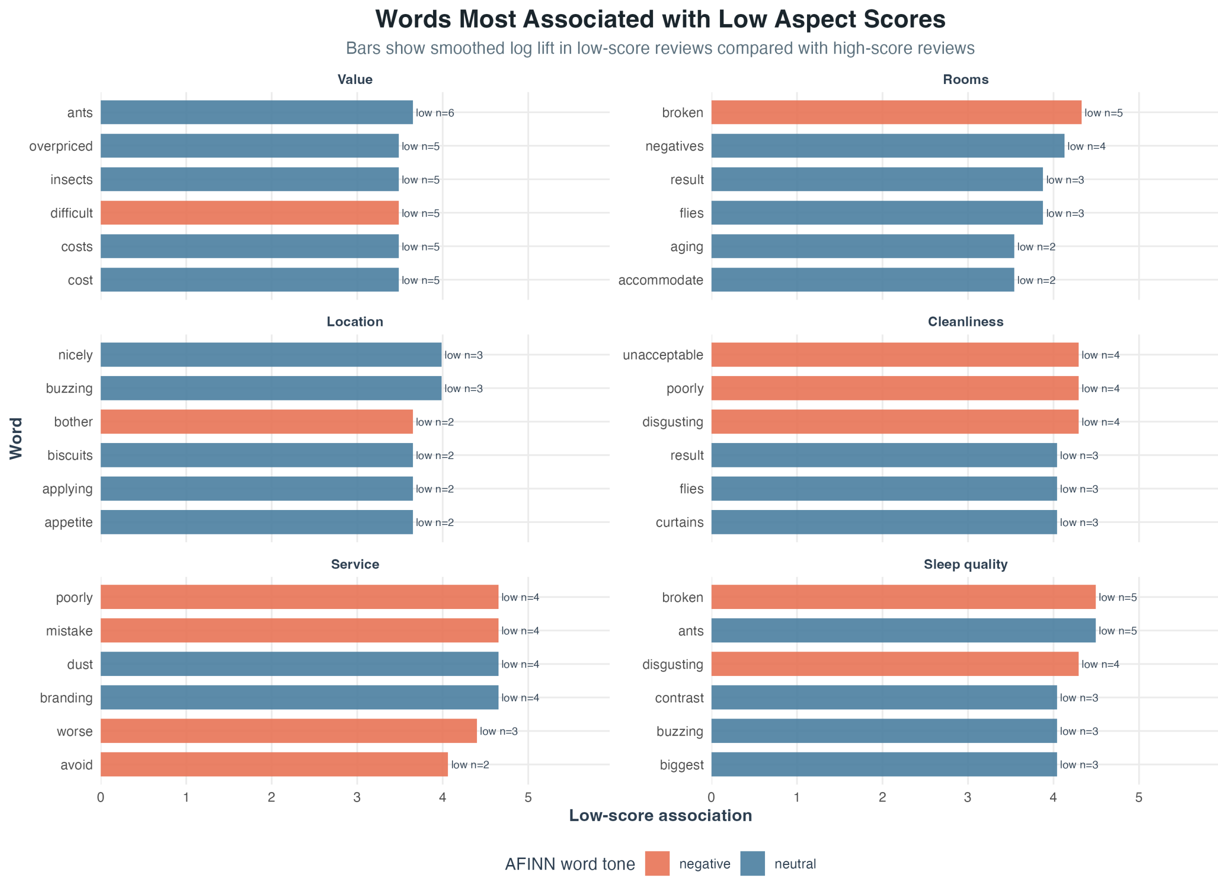

Figure 9. Terms appearing unusually often in low-aspect reviews.

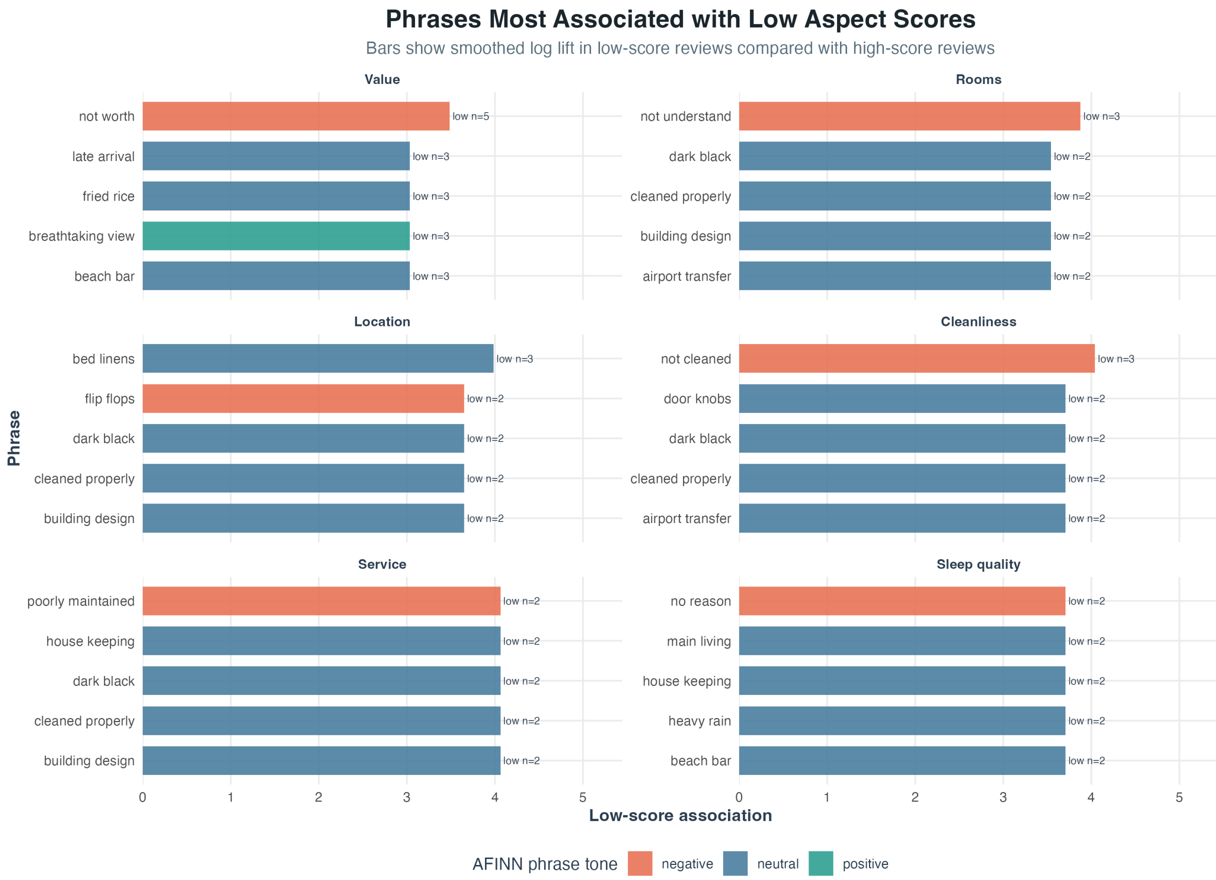

Figure 10. Two-word phrases appearing unusually often in low-aspect reviews.

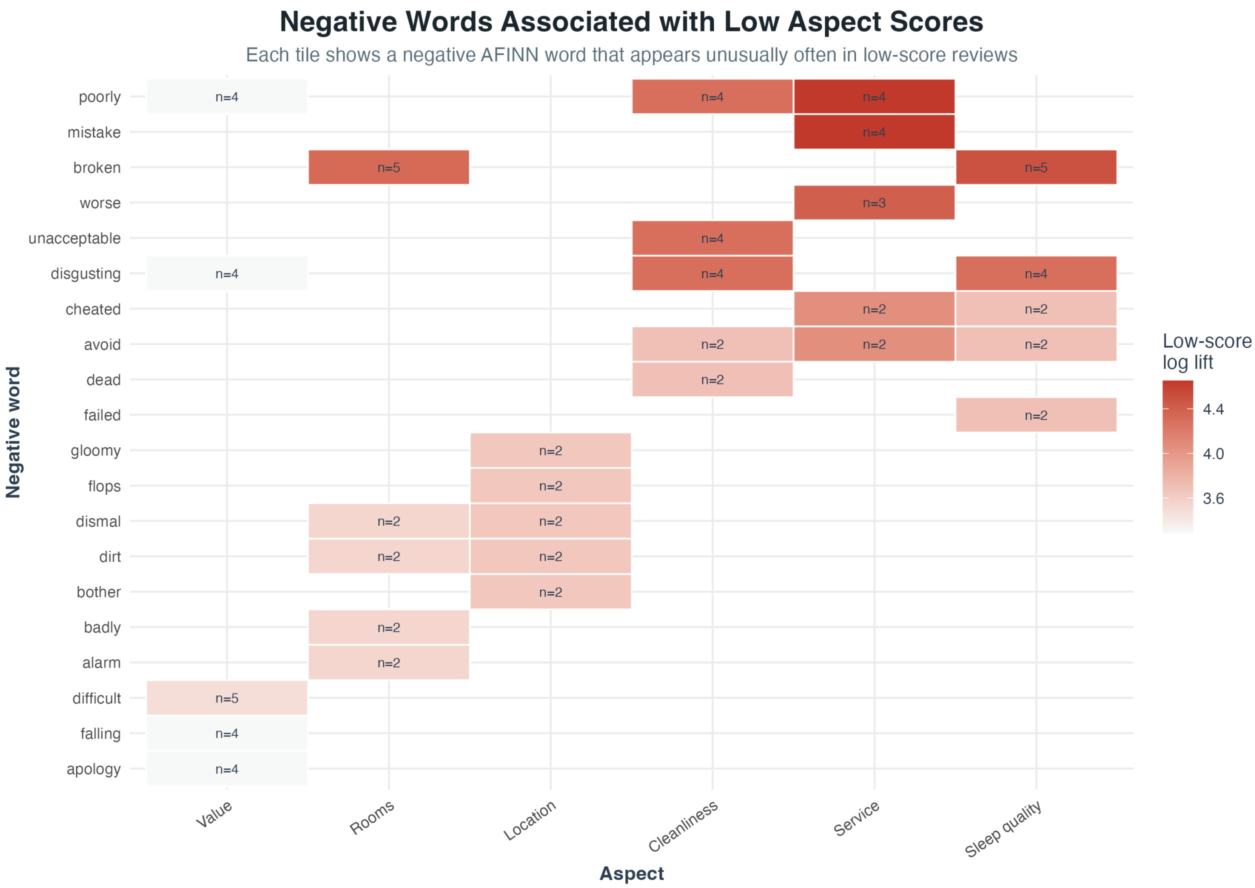

Figure 11. Negative sentiment terms associated with low-aspect review text.

The mismatch table adds another layer of interpretation. `aspect_text_mismatches.csv` currently contains 320 aspect-review rows with at least one signal disagreement. Because a single row can have more than one mismatch type, the categories overlap: 205 rows have low aspect ratings with positive text sentiment, 100 rows have low overall ratings with high aspect ratings, 80 rows have high overall ratings with low aspect ratings, and 37 rows have high aspect ratings with negative text sentiment. The largest mismatch counts appear in Value (78), Location (61), Service (54), Rooms (51), Cleanliness (42), and Sleep quality (34). These cases should be read qualitatively because they often contain the nuance that aggregate averages hide.

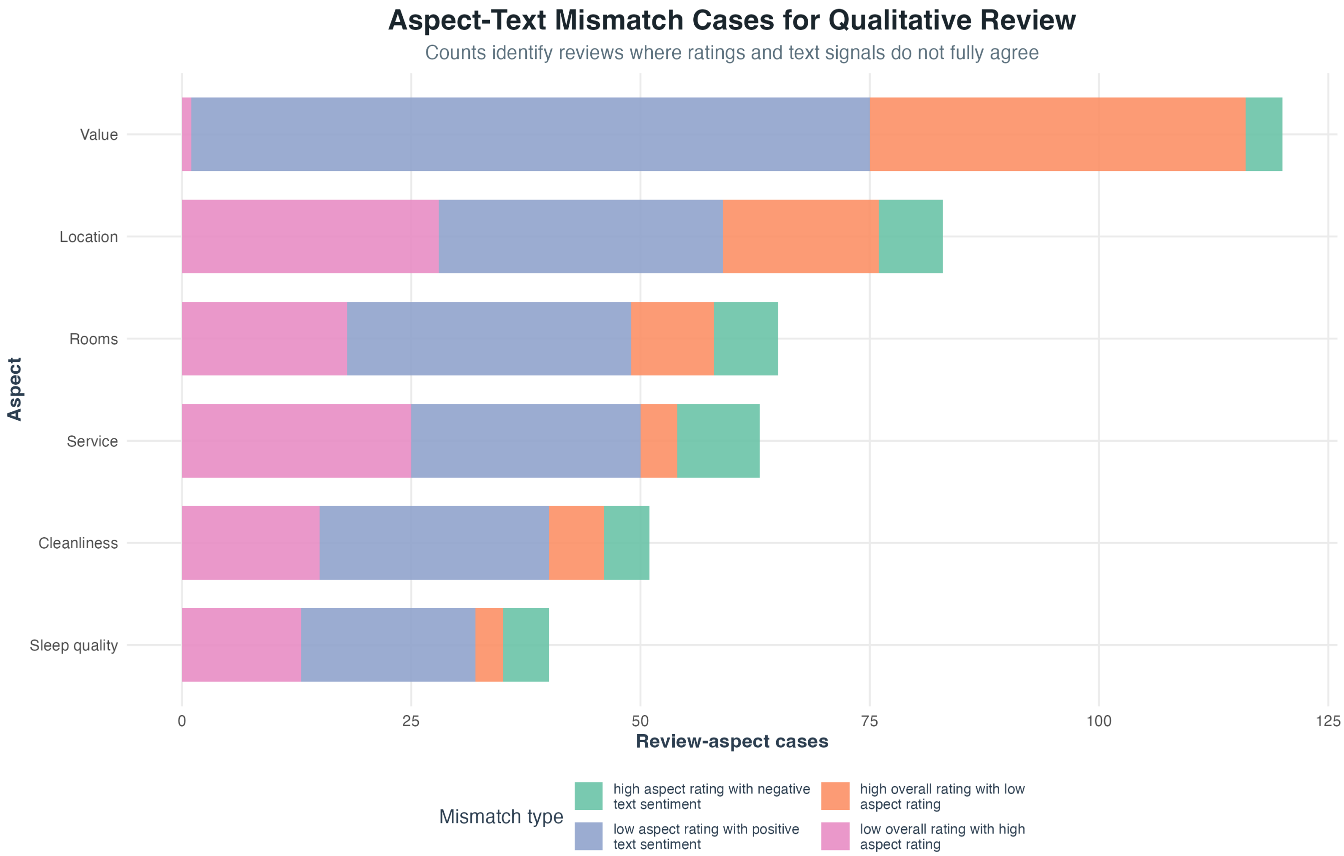

Figure 12. Aspect-text mismatch counts by aspect and mismatch type.

The interpretation remains cautious. Aspect ratings apply to whole reviews, not individual sentences. Therefore, a word associated with low service ratings should be described as service-associated text, not as definitive evidence that the word refers only to service. This distinction is important for academic validity and for responsible managerial use because automated sentiment features can identify patterns but do not by themselves establish causality or operational root cause (Liu, 2020).

11. Temporal Analysis

11.1 Date Parsing

The visualization script includes a `parse_review_dates()` helper that attempts several common date formats:

- year-month-day
- month-day-year
- day-month-year
- month-year

This is necessary because review datasets often contain inconsistent date formats. Reviews with unparseable dates are excluded from time-series visualization but retained in the scored dataset.

11.2 Monthly, Quarterly, and Yearly Aggregation

The methodology uses multiple period lengths because each answers a different question:

Period	Purpose
Monthly	Detect short-term changes and seasonal signals.
Quarterly	Reduce monthly noise and support operational review cycles.
Yearly	Summarize long-run distribution and structural change.

The project generates:

- Monthly sentiment heatmap.
- Monthly rolling sentiment trend.
- Quarterly sentiment distribution.
- Yearly sentiment distribution.

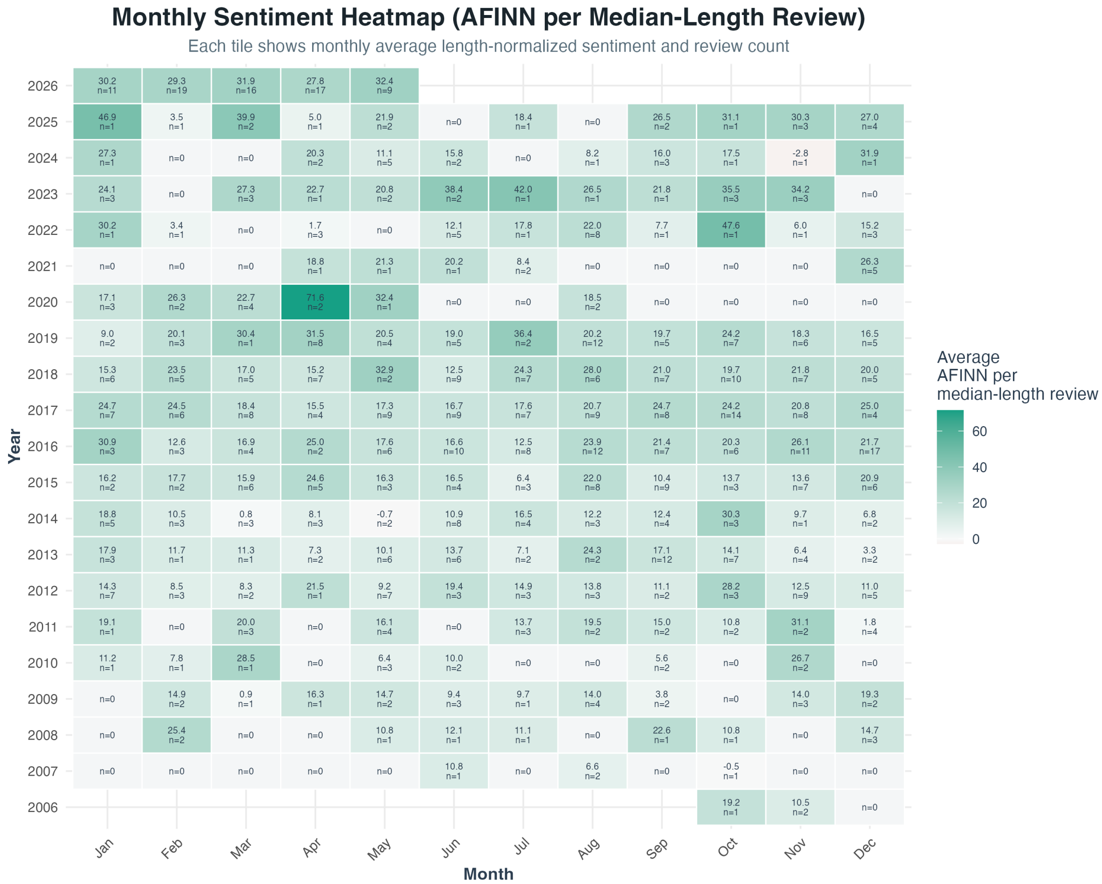

Figure 13. Monthly median-length-normalized sentiment heatmap.

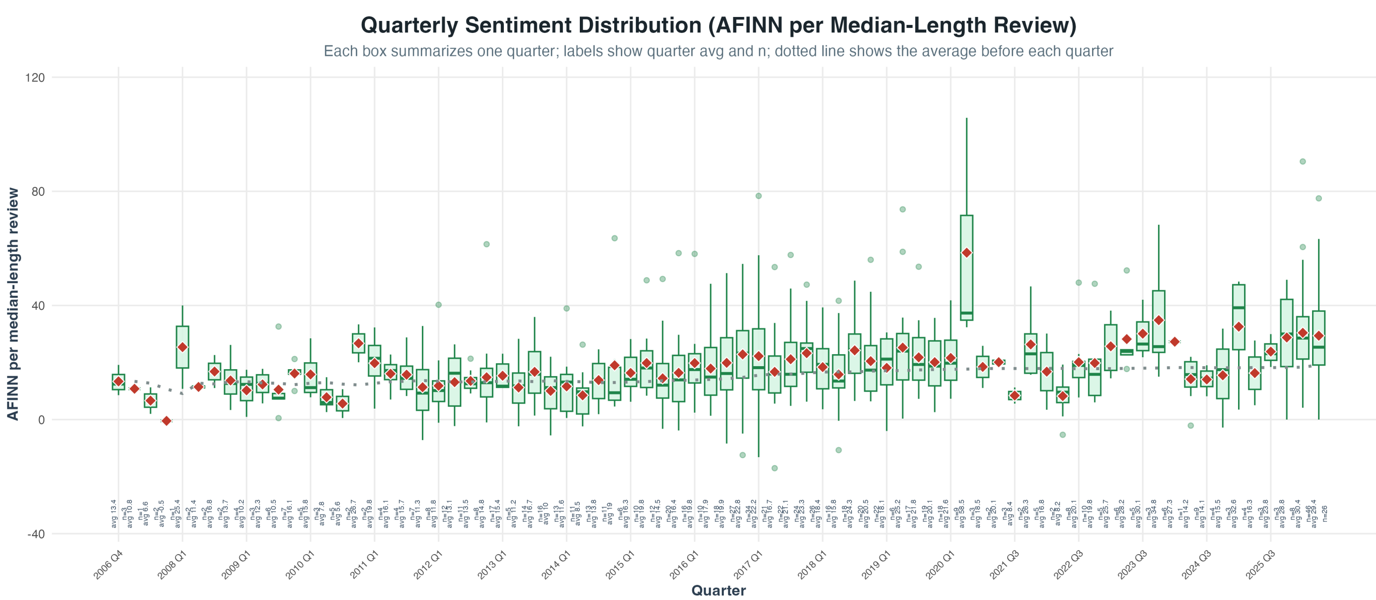

Figure 14. Quarterly distribution of median-length-normalized AFINN sentiment.

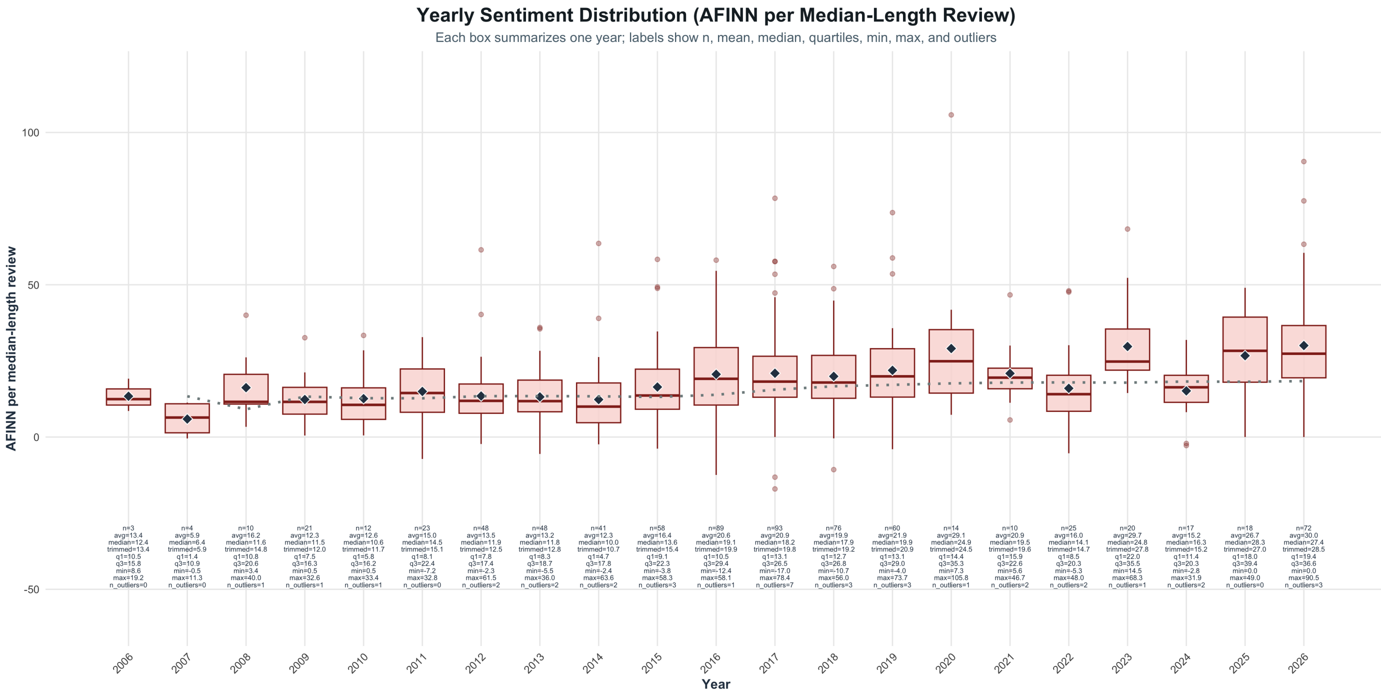

Figure 15. Yearly distribution of median-length-normalized AFINN sentiment.

11.3 Rolling Average and Prior Average

The monthly rolling chart uses a six-month rolling average. This smooths noisy month-level changes, especially when some months have few reviews. The script also computes a prior average, defined as the average sentiment before the current period. This avoids comparing a period to a future-informed all-time average. If decision-support functionality is later implemented, prior baselines would be more realistic because managers only know past performance at the time a decision is made. The need for cautious temporal interpretation is reinforced by research showing that platform context and cultural review behavior can affect TripAdvisor hotel-review patterns (Filieri et al., 2015; Litvin, 2019).

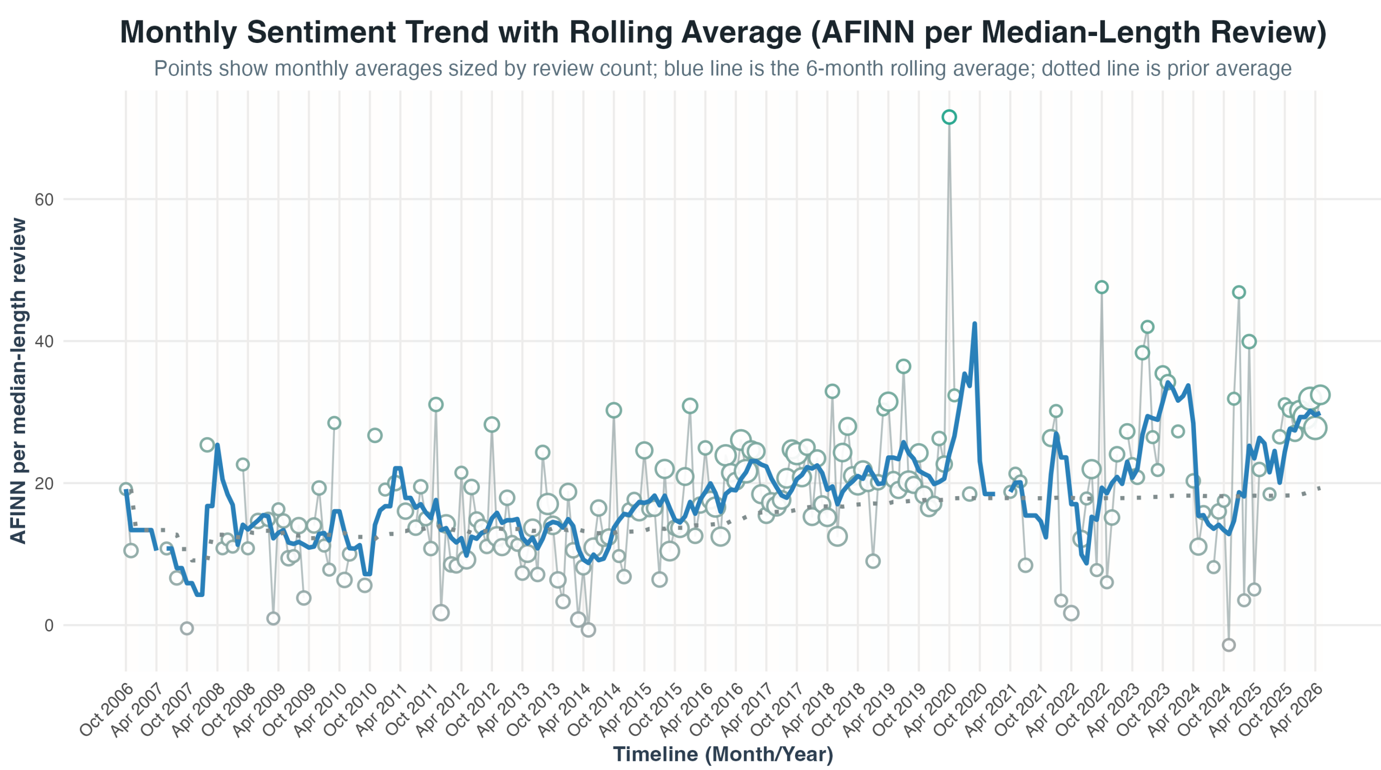

Figure 16. Monthly sentiment points, six-month rolling average, and prior-average baseline.

12. Statistical Drift Monitoring and Future Decision Support

12.1 Rationale

The current project now implements statistical drift-monitoring outputs, but it does not implement management recommendations or investment recommendations. This distinction matters. The code can show when a period looks unusual relative to prior review history, but it does not decide what the hotel should do. Operational action still requires qualitative reading, business context, and managerial judgment.

In operational terms, the future decision-support question remains:

If the rating or sentiment of reviews drifts away from the mean, median, or historical baseline over a specified number of periods, what should management do?

The current implementation answers the narrower methodological question: which periods deserve closer inspection because their sentiment or rating distribution is unusually low compared with earlier reviews?

12.2 Implemented Period Monitoring Metrics

The visualization script writes `output/reports/sentiment_period_summary.csv`. For each month, quarter, and year, it reports:

- Mean star rating.
- Median star rating.
- Trimmed mean star rating.
- Mean, median, and trimmed mean AFINN sentiment on the raw summed scale.
- Mean, median, and trimmed mean AFINN sentiment after scaling each review to the dataset's median cleaned review length.
- Mean, median, and trimmed mean Syuzhet sentiment on both the raw and median-length-normalized scales.
- Percentage of reviews rated 1 to 3.
- Percentage of reviews with negative median-length-normalized AFINN sentiment.
- Review count for the period.
- Historical median and historical MAD for prior-period AFINN and rating baselines.
- Robust z-score fields for median sentiment and median rating.
- Statistical drift flags for unusually low sentiment or rating periods.

The review count is treated as a required confidence input. A period with two reviews should not be interpreted the same way as a period with fifty reviews unless the content indicates a severe risk.

The current `sentiment_period_summary.csv` contains 286 month, quarter, and year rows. Under the implemented minimum-volume and robust-z rules, no period is currently flagged as a statistical drift period. This should be interpreted narrowly: it means the coded statistical monitor did not find a period that crossed its current robust threshold, not that the property has no operational issues. The aspect-text and qualitative-review outputs still identify specific reviews and aspect dimensions that deserve manual interpretation.

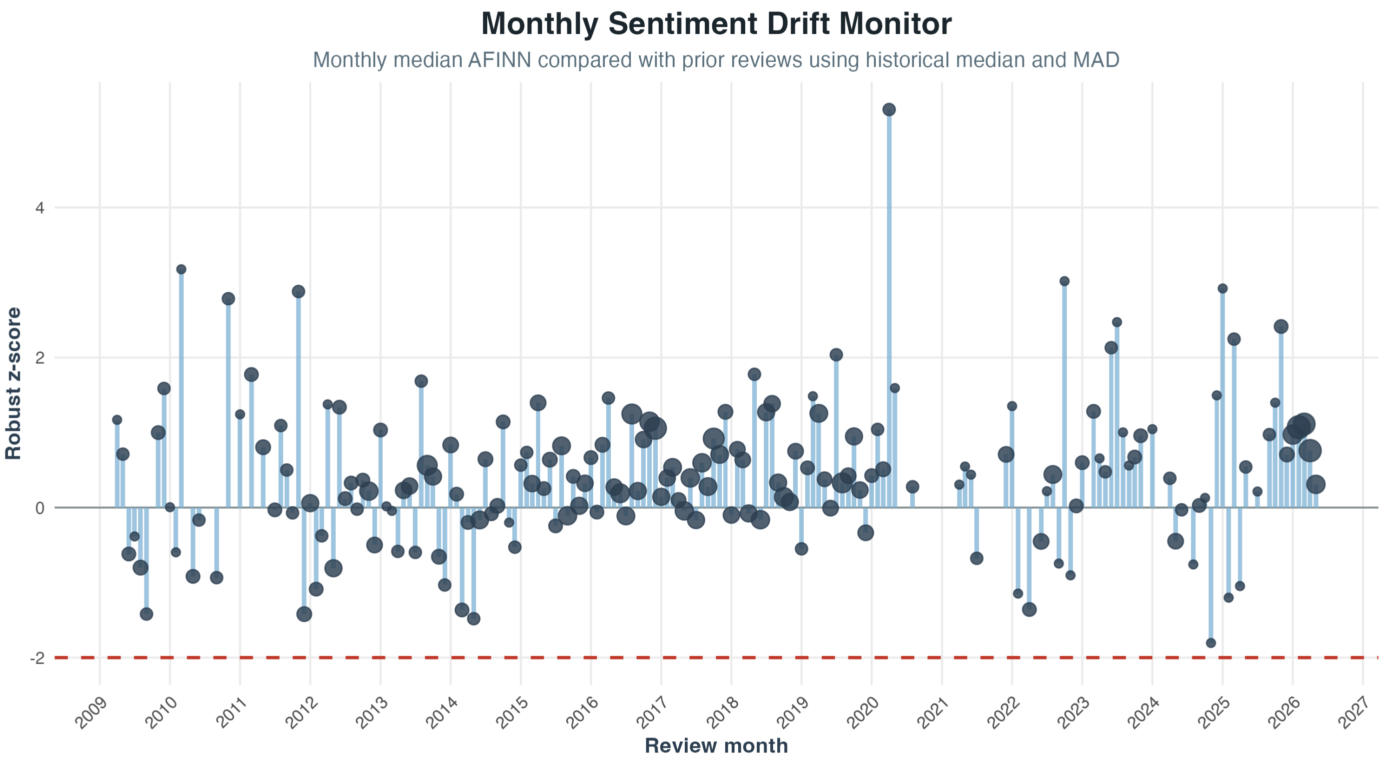

Figure 17. Monthly robust sentiment-drift monitor using prior review history.

12.3 Baseline Definitions

The implemented baseline is a prior cumulative baseline: each period is compared only with reviews posted before that period. This avoids future leakage. For example, a month in 2024 is not compared with reviews from 2025 or 2026.

Future decision-support work could add two additional baselines:

- Rolling baseline: the mean or median over the previous k periods.
- Seasonal baseline: the same month or quarter in prior years.

For luxury hospitality, a rolling baseline is likely to be the most practical. A six-month or twelve-month rolling baseline could detect current drift without being dominated by early historical reviews. Seasonal baselines would be useful when travel demand and guest mix are seasonal.

12.4 Robust Drift Calculation

For a monitored metric M in period t, simple drift is:

```
drift_t = M_t - baseline_t
```

For metrics where lower is worse, such as average rating or average sentiment, negative drift would be the concern. For metrics where higher is worse, such as low-rating share or anger count, positive drift would be the concern.

The implemented monitor uses a robust standardized form for median sentiment and median rating:

```
robust_z_t = (period_median_t - historical_median_t) / historical_MAD_t
```

MAD means median absolute deviation. It is preferable here because review sentiment is skewed, review volume varies by period, and isolated outlier reviews should not control the baseline. The script only calculates robust z-scores after enough prior reviews exist. It sets a statistical flag when the current period has at least three reviews, the prior baseline has at least twenty reviews, and the robust z-score is at or below -2. This flag is a triage signal, not proof of operational failure.

12.5 Future Persistence Rule

A future framework should not react to every fluctuation. It should evaluate persistence over n periods:

```
trigger if drift is unfavorable for n consecutive periods
```

For a future decision-support extension, the following default ladder is recommended:

Level	Trigger	Interpretation
Watch	Metric is worse than baseline for 2 consecutive periods.	Possible emerging issue.
Investigate	Metric is worse than baseline by more than normal variation for 2 periods.	Likely operational signal.
Act	Low-rating share doubles or sentiment declines for 3 consecutive periods.	Intervention required.
Escalate	Service, cleanliness, rooms, safety, or value drops sharply with negative text evidence.	Senior management and investment review required.

The number of periods should be adjusted based on review volume. If monthly review volume is low, quarterly periods would be more reliable.

12.6 Minimum Volume Rule

The current script uses minimum counts before assigning statistical flags, but it does not yet assign interpretive confidence labels. A future decision-support implementation should assign each period a confidence label:

Period review count	Confidence
1 to 4	Low confidence; read reviews manually.
5 to 9	Moderate confidence; combine with adjacent periods if needed.
10 or more	Higher confidence for aggregate metrics.

This would prevent overreaction to thin data. However, severe complaints involving safety, cleanliness, discrimination, or major service failure should still be reviewed immediately even if the period count is small.

13. Future Work: Pattern-to-Action Interpretation

Pattern-to-action mapping is not currently implemented in the repository. It is a proposed future layer that could be developed after the statistical drift outputs are validated and trigger-state rules are defined. The current drift metrics become managerially useful only when different patterns are mapped to different action pathways. This proposed direction is grounded in hospitality research linking service quality, image, and guest experience to satisfaction, loyalty, and revisit behavior (Bichler et al., 2021; Kandampully & Suhartanto, 2000).

Observed pattern	Likely interpretation	Illustrative management response	Illustrative investment response
Rating down and sentiment down	Broad guest-experience decline.	Root-cause review, department-level service audit, immediate service recovery.	Consider targeted investment if text points to assets, rooms, F&B, or facilities.
Rating stable but sentiment down	Guests still rate highly, but language is cooling.	Early warning review; inspect high-rating reviews with negative text.	Protect experience drivers before ratings decline.
Rating down but sentiment stable	Value or expectation mismatch.	Review price communication, package design, inclusions, and pre-arrival promises.	Invest in perceived-value enhancements before discounting.
Value rating down	Price no longer feels justified.	Audit inclusions, transparency, upsell friction, and guest expectation setting.	Invest in visible value creators or reposition packages.
Service rating down	Core luxury promise is weakening.	Retrain staff, audit butler handoffs, review staffing ratios, improve recovery protocols.	Invest in staffing, language capability, training, and retention.
Rooms or cleanliness down	Physical product or maintenance issue.	Inspect recurring room/villa defects and housekeeping process.	Prioritize capex, preventive maintenance, refurbishment, or quality-control systems.
Food and beverage complaints rise	Dining experience is weakening.	Review menu, breakfast, wait times, price-value, and service coordination.	Invest in kitchen/service capacity, restaurant refresh, or menu development.
Negative emotions increase	Emotional intensity of complaints is rising.	Read negative-emotion reviews manually; classify by failure type.	Escalate if linked to structural experience gaps.
Sentiment volatility rises	Experience consistency problem.	Compare peak/off-peak periods, staff shifts, occupancy, and guest segments.	Invest in process standardization and operational resilience.

14. Future Work: Qualitative Review of Outliers

When the statistical drift monitor flags a period, it should always be followed by qualitative reading. The recommended future process is:

- Identify periods or segments that triggered watch, investigate, act, or escalate.

- Extract reviews from those periods with low rating, low sentiment, high anger, high disgust, or high negative word counts.
- Read the original `review_text`, not only the cleaned text.
- Code the complaint themes manually or semi-manually.
- Distinguish controllable issues from uncontrollable factors.
- Assign each controllable theme to an operational owner.
- Track whether the theme persists after intervention.

This qualitative step would protect the method from false precision. A negative score can show where to look, but the review text explains what actually happened. This follows the interpretive logic of the reference TripAdvisor sentiment study, which combines lexical sentiment results with qualitative reading of review language and context (Adnyana et al., 2026).

15. Visualization Strategy

The project generates several visual outputs, each with a methodological purpose:

Figure	Purpose
Sentiment distribution	Shows overall positive, neutral, and negative classification balance.
Rating-sentiment boxplot	Compares textual sentiment with star ratings and identifies divergence.
Country sentiment bar chart	Compares median length-normalized sentiment across reviewer countries with enough visible reviews.
Aspect mean ratings	Compares structured value, rooms, location, cleanliness, service, and sleep-quality ratings.
Aspect low-score share	Identifies dimensions where 1-3 star aspect ratings are concentrated.
Yearly aspect-rating heatmap	Shows whether structured aspect ratings change across years while displaying available rating counts.
Aspect sentiment boxplot	Compares whole-review text sentiment across structured aspect-rating levels.
Aspect low-score key terms	Shows words most associated with low aspect scores compared with high aspect scores.
Aspect low-score key phrases	Shows two-word phrases most associated with low aspect scores compared with high aspect scores.
Aspect negative-term heatmap	Highlights negative sentiment words that appear unusually often in low-aspect reviews.
Aspect-text mismatch counts	Shows where rating and text signals disagree and require qualitative review.
Emotion breakdown	Shows dominant NRC emotion categories across the corpus.
Guest experience word cloud	Combines sentiment/emotion words with hotel-experience and frequent context terms after filtering broad filler words.
Monthly heatmap	Shows median-length-normalized sentiment intensity by month and year.
Monthly rolling trend	Shows smoothed median-length-normalized sentiment movement and prior-average comparison.
Monthly drift monitor	Shows monthly robust z-scores for median sentiment against prior review history.
Quarterly boxplot	Shows within-quarter distribution of AFINN scaled to a median-length review.
Yearly boxplot	Shows long-term distribution, average, median, trimmed mean, quartiles, and outliers for AFINN scaled to a median-length review.

Visualizations should be interpreted as analytical aids, not as evidence by themselves. Each chart should be connected back to the underlying reviews and operational context.

16. Reliability, Validity, and Triangulation

16.1 Reliability

Reliability is supported by scripted preprocessing, centralized configuration, deterministic cleaning rules, and reproducible outputs. The same raw CSV and scripts should produce the same cleaned and scored files. The use of tidy data principles also supports repeatable text-mining workflows because each token, review, and score is represented in an inspectable tabular form (Silge & Robinson, 2016).

16.2 Construct Validity

Construct validity is strengthened by measuring guest experience through multiple indicators, which reduces dependence on any single sentiment score or rating field:

- Star rating.
- Textual sentiment.
- Emotion categories.

- Aspect ratings.
- Review text themes.
- Temporal patterns.

No single indicator is treated as sufficient. For example, a decline in average AFINN score should be checked against star ratings, aspect ratings, and review narratives.

16.3 Internal Validation

Internal validation is achieved through triangulation across lexicons and structured ratings. If ratings, AFINN scores, Syuzhet scores, and negative-emotion counts all move in the same direction, confidence increases. If they diverge, the divergence itself becomes an analytical finding. This triangulation is appropriate because sentiment-analysis methods are useful but imperfect proxies for human opinion and emotion (Liu, 2020).

16.4 External Validity

External validity is limited because the study focuses on one luxury resort and one review platform. The analytical workflow can be adapted to other hotels, but any future drift thresholds and interpretation rules should be recalibrated for each property's review volume, market segment, language mix, and brand promise. This caution is consistent with evidence that TripAdvisor hotel reviews are shaped by traveler trust, platform behavior, and cultural differences in review expression (Filieri et al., 2015; Litvin, 2019).

17. Ethical and Data Governance Considerations

The project uses online review data, but public availability does not remove ethical responsibility. Because TripAdvisor reviews can influence traveler trust and business reputation, review analytics should report aggregate evidence carefully and avoid unnecessary person-level disclosure (Filieri et al., 2015). The following safeguards are recommended:

- Report aggregate patterns rather than identifying individual reviewers.
- Avoid quoting long review passages unless necessary and properly attributed.
- Do not track reviewer names in this project; if future work requires identifiable reviewer information, keep it outside the public analysis outputs and document the access controls.
- Preserve raw text for auditability but limit unnecessary redistribution.
- Avoid using sentiment scores to profile or target individual guests.
- If decision-support features are later added, treat algorithmic outputs as supporting evidence, not as final judgments.
- Respect platform rules and data-use restrictions when acquiring or refreshing data.

The ethical stance is particularly important because luxury-hotel reviews may include personal travel details, family arrangements, special occasions, complaints, and staff names.

18. Limitations

Several limitations must be acknowledged.

First, TripAdvisor reviewers are self-selected. They may not represent all guests, and review behavior may vary by nationality, age, travel purpose, satisfaction intensity, cultural background, and digital habits (Filieri et al., 2015; Litvin, 2019).

Second, star ratings are skewed toward positive evaluations. In this corpus, the median rating is 5.0, which means small declines in the average may matter even when the overall rating remains high.

Third, lexicon-based sentiment analysis can misinterpret context. The implemented scoring now handles simple local negation windows, but it may still struggle with sarcasm, complex negation, multilingual content, idioms, local cultural terms, and luxury-hospitality vocabulary (Liu, 2020).

Fourth, summed AFINN and Syuzhet scores are influenced by review length. Longer reviews may receive larger positive or negative totals simply because they contain more sentiment words. The trend charts and aspect-text diagnostic summaries reduce this problem by scaling sentiment to the dataset's median cleaned review length where period or aspect means are compared. Any remaining displays that use summed sentiment scores should still be read with review length in mind.

Fifth, review dates may not equal stay dates. A guest may post a review weeks or months after the actual stay. Where possible, stay dates should be used for operational diagnosis and review dates should be used for public-perception timing.

Sixth, platform and scraping conditions may affect what data are available. Historical reviews, hidden text, management responses, translated reviews, or deleted reviews may not be consistently captured.

Seventh, investment decisions require financial data not present in the review corpus. Sentiment can identify demand-side pain points and experience drivers, but it cannot by itself estimate return on investment. A future investment layer would need to combine review analytics with occupancy, ADR, RevPAR, margin, capex cost, maintenance logs, complaint records, and competitive benchmarking. The link from guest experience to business outcomes is plausible in hospitality research, but the financial inference would require data beyond text reviews (Kandampully & Suhartanto, 2000).

19. Recommended Extensions

The current methodology can be extended in several ways:

- Compare raw and median-length-normalized AFINN and Syuzhet side by side in selected diagnostic tables so readers can distinguish total evaluative content from normalized sentiment intensity.
- Extend the current aspect-text analysis from review-level weak labels to sentence-level or phrase-level aspect classification for service, rooms, value, dining, spa, beach, and privacy.
- Use manual coding on low-rating and high-negative-emotion reviews to validate automated themes.
- Add multilingual handling for non-English reviews.
- Compare review-date trends with stay-date trends.
- Extend the implemented drift metrics to low-rating share, negative-share, emotion rates, and aspect ratings.
- Add trigger-state logic for watch, investigate, act, and escalate categories.
- Add a qualitative review workflow for reviews that trigger low-rating, low-sentiment, or high-negative-emotion alerts.
- Incorporate management responses and response time as service recovery indicators.
- Add competitor benchmarks for comparable luxury resorts in Bali.
- Link sentiment drift to occupancy, ADR, RevPAR, renovation timelines, maintenance records, and complaint logs.
- Build a dashboard with trigger states and action recommendations.
- Conduct sensitivity analysis using monthly, quarterly, and semiannual periods.
- Add unsupervised review-authenticity risk screening as a future methodological extension, using reviewer sparsity, text-shape anomalies, rating-sentiment mismatch, near-duplicate detection, temporal context, and multivariate anomaly models to prioritize manual inspection rather than classify reviews as fake. This extension should explicitly separate period-level signals from review-level evidence: high review volume in a month should trigger contextual review, not automatically increase suspicion for every review in that month. Robustness checks should include temporal ablation, threshold sensitivity analysis, and human validation of sampled low-, moderate-, and high-risk reviews.

20. Conclusion

This methodology transforms TripAdvisor reviews of Bvlgari Resort Bali into a structured research dataset and visualization workflow. It begins with data validation and text preprocessing, applies multiple lexicon-based sentiment and emotion measures, compares text sentiment with star and aspect ratings, visualizes sentiment patterns over time, and produces robust period-monitoring outputs. The approach builds on established review-based hospitality and sentiment-analysis methods while adapting them to a single luxury-resort case (Adnyana et al., 2026; Filieri et al., 2015; Liu, 2020). The current project does not yet implement management or investment decision logic. Its future methodological contribution would be a decision-support extension in which persistent unfavorable drift is mapped to tiered management and investment responses.

The framework is academically rigorous because it specifies the corpus, preprocessing rules, analytical measures, validation logic, limitations, and ethical constraints. It is also a foundation for future managerial use because it produces the indicators that a later decision-support system would need. The central principle is caution with actionability: do not treat sentiment movement as meaningful until it is persistent, supported by enough review volume, triangulated with ratings or aspect scores, interpreted through the actual review text, and connected to operational or financial evidence.

References

Adnyana, P. P., Wiweka, K., Lochan, A., & Trisdyani, N. L. P. (2026). Beyond taste: A sentiment analysis of informal language and cultural appreciation in Tripadvisor reviews of Balinese Ayam Betutu. *SOSIOHUMANIORA: Jurnal Ilmiah Ilmu Sosial dan Humaniora, 12*(1), 176-197. https://doi.org/10.30738/sosio.v12i1.21041

Bichler, B. F., Pikkemaat, B., & Peters, M. (2021). Exploring the role of service quality, atmosphere and food for revisits in restaurants by using a e-mystery guest approach. *Journal of Hospitality and Tourism Insights, 4*(3), 351-369. https://doi.org/10.1108/JHTI-04-2020-0048

Filieri, R., Alguezaui, S., & McLeay, F. (2015). Why do travelers trust TripAdvisor? Antecedents of trust towards consumer-generated media and its influence on recommendation adoption and word of mouth. *Tourism Management, 51*, 174-185. https://doi.org/10.1016/j.tourman.2015.05.007

Kandampully, J., & Suhartanto, D. (2000). Customer loyalty in the hotel industry: The role of customer satisfaction and image. *International Journal of Contemporary Hospitality Management, 12*(6), 346-351. https://doi.org/10.1108/09596110010342559

Litvin, S. W. (2019). Hofstede, cultural differences, and TripAdvisor hotel reviews. *International Journal of Tourism Research, 21*(5), 712-717. https://doi.org/10.1002/jtr.2298

Liu, B. (2020). *Sentiment analysis: Mining opinions, sentiments, and emotions* (2nd ed.). Cambridge University Press. https://doi.org/10.1017/9781108639286

Mohammad, S. M., & Turney, P. D. (2013). *NRC Emotion Lexicon*. National Research Council Canada. https://doi.org/10.4224/21270984

Nielsen, F. A. (2011). A new ANEW: Evaluation of a word list for sentiment analysis in microblogs. *Proceedings of the ESWC2011 Workshop on Making Sense of Microposts*, 93-98. https://doi.org/10.48550/arXiv.1103.2903

Silge, J., & Robinson, D. (2016). tidytext: Text mining and analysis using tidy data principles in R. *Journal of Open Source Software, 1*(3), 37. https://doi.org/10.21105/joss.00037